

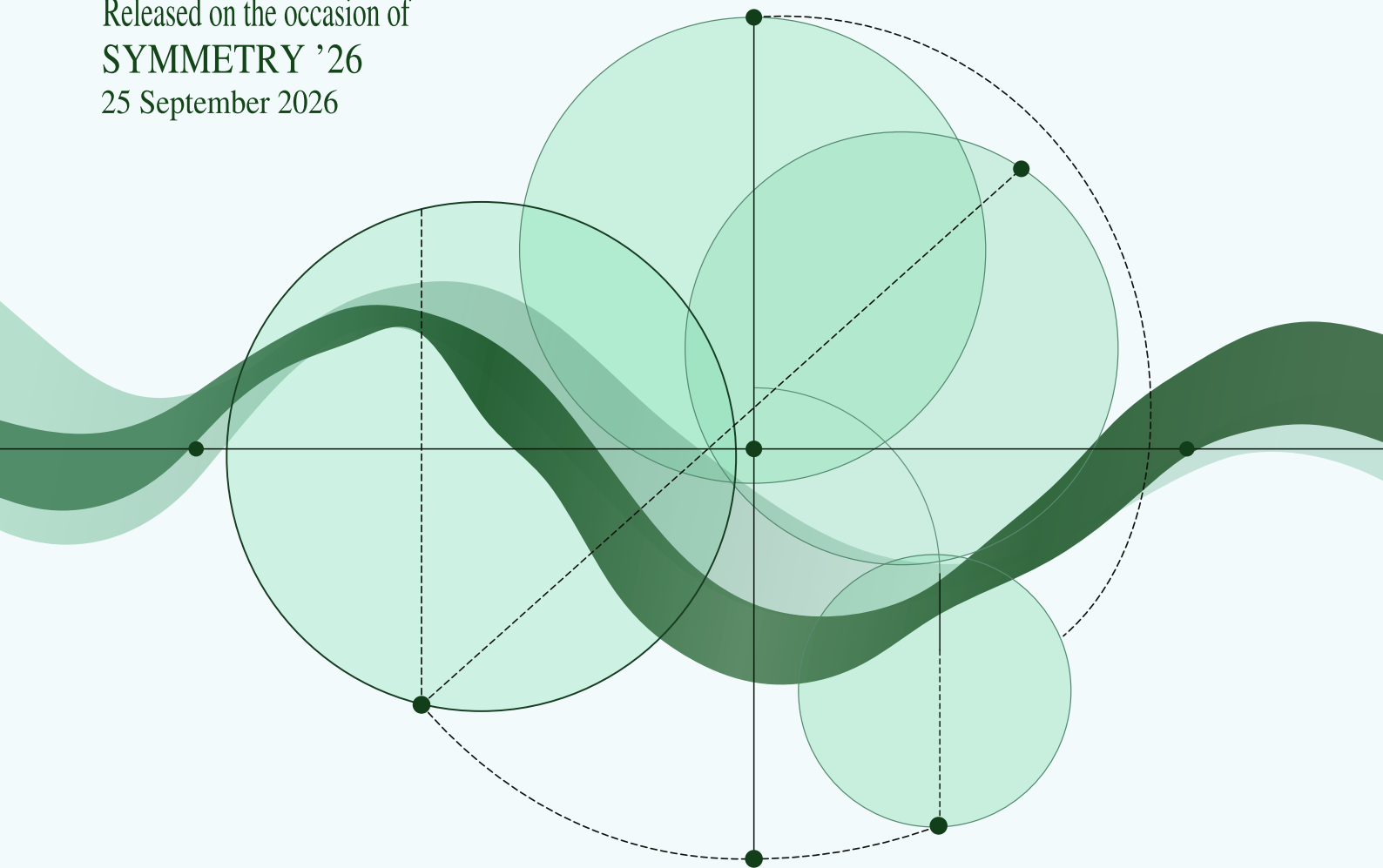
DEPARTMENT OF MATHEMATICS
PRESIDENCY UNIVERSITY, KOLKATA

elements

• VOLUME 1 •

Mathematics Departmental Magazine

Inaugural Issue
Released on the occasion of
SYMMETRY '26
25 September 2026



“Mathematics is the music of reason”

— James Joseph Sylvester

Message from the Head of the Department



Dr. Avishek Adhikari

Professor and Head
Department of Mathematics
Presidency University, Kolkata

Welcome to the inaugural issue of *Elements*.

It is with immense pride and a deep sense of anticipation that I write this message for the very first edition of our departmental magazine. The launch of *Elements* marks a significant milestone in the life of our department. It is a moment that celebrates not just the culmination of our academic endeavors, but the very spirit of curiosity and inquiry that defines us.

Why *Elements*? In mathematics, the elements are the fundamental building blocks from which everything complex is derived. From the simplest axioms and integers, we build the magnificent and intricate structures of calculus, algebra, and topology. This magazine is a testament to that process—a collection of the “elements” of our department: the fundamental units of thought, discovery, and passion that our students and faculty bring to the world of numbers.

I extend my deepest gratitude to the editorial board for their vision and tireless effort in bringing this inaugural issue to life. To the contributors, thank you for sharing your insights and your voice. And to our readers, I hope you find within these pages the same spark of wonder that has driven mathematicians for millennia.

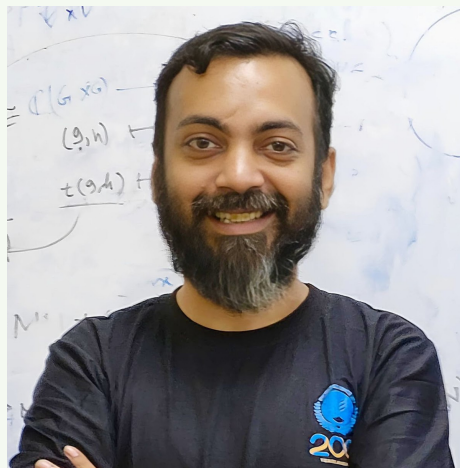
May this first issue of *Elements* be the first of many. On behalf of the entire department, I wish *Symmetry* '26 a resounding success.

With warm regards,

Dr. Avishek Adhikari

Professor and Head, Department of Mathematics
Presidency University, Kolkata

Message from the Editor



Dr. Angsuman Das

Assistant Professor
Department of Mathematics
Presidency University, Kolkata

Dear Readers,

It gives me great pleasure to present to you the very first issue of *Elements*. What began as a casual conversation among a few of us has now taken shape as a magazine that we hope will become a lasting tradition in our department.

This magazine is, above all, a platform—a space where ideas can be shared freely, where curiosity is celebrated, and where the mathematical community within our department can come together. Whether you are a first-year student encountering the beauty of proofs for the first time, or a senior delving into research-level problems, there is something here for you.

Putting this issue together has been a labor of love. I would like to thank every contributor who trusted us with their work, the faculty who supported us at every step, and my fellow student editors whose dedication made this possible.

As you turn these pages, I invite you to read with an open mind. Mathematics, after all, is not just a subject to be studied—it is a way of seeing the world.

Happy reading!

Dr. Angsuman Das
Editor, *Elements*

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“The mathematician’s patterns, like the painter’s or the poet’s, must be beautiful.”
— G. H. Hardy

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Articles

CHAPTER 1

Why Your Clock Still Speaks Babylonian

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Look around, look at your clock, your phone or whatever ticking device is nearest to you, you see the recurring number—60 seconds, 60 minutes, 12 hours—these are not some random magic numbers that shape your entire day. Every time you check the time, you’re doing math the way people used to do over 4,000 years ago, in a civilisation before the mighty Rome was even founded.

It Started With a Hand, Not a Decision

It didn’t start with a decision, it started with a “hand”. Base-60 didn’t even start with the Babylonians. They inherited it from the Sumerians, who were doing accounting in Mesopotamia as early as 3000 BCE, over a thousand years before Babylon became a major power. The leading theory about how “60” came into this picture is not some elegant mathematical insight, it’s a simple hand trick.

If you use your thumb to count the three knuckle segments of your other four fingers, you get 12, and if you keep doing the same to see how many “sets of 12” you’ve completed, 12×5 gets you 60 with no writing required. This did not originate on some whiteboard, it fell out of people who were already counting on their hands, labours and merchants for quick trade in the market. Another theory suggests that the Sumerians deliberately chose 60 because it’s divisible by 1, 2, 3, 4, 5, 6 as well as 10, 12, 15, 20 and 30. Historians believe both contributed to the development.

Accounting for Grain, Not for Glory

The Babylonians and Sumerians weren’t developing this number system for the sake of pursuing “Mathematics” as an abstract subject. Base-60 shows up primarily and most heavily in administrative record keeping like rationing grains, dividing land, calculating debt. The math that eventually structured how the modern world tells the time and measures angles was invented to keep warehouse paperwork tidy.

From the Granary to the Sky

It gets more interesting because Babylonian astronomy grew directly out of this same bureaucratic mindset, just aimed upward instead of at a granary. Priest-scribes tracked the moon, sun, and planets obsessively, not from scientific curiosity in our modern sense, but because celestial events were read as omens relevant to the King and the state. And because these scribes were fluent in the base-60 arithmetic form from their day jobs, it became the obvious tool to build the mathematical models required for the cause. Surprisingly, it all fit all too well. A circle divided into 360 degrees beautifully with sexagesimal math. Some historians link 360 with the Babylonian approximation

that a year has around 360 days. Although the link is debatable, the bigger point isn't: an accounting methodology invented for business purposes advocated its way into the "heavens", turning out to be exactly the right tool for splitting up circles, tracking cycles, plotting angles.

It worked so well that, centuries later, Greek astronomers, including Ptolemy, adopted the Babylonian sexagesimal system for their astronomical tables, even while using entirely different numerals for everyday counting. Math and stargazing joined forces to reveal the far-fetched secrets of the universe.

The Two Holdouts: Time and Angles

The world eventually switched to base-10 except in two places: time and angles. Decimal is easier to *learn* while base-60 is better at *dividing*, and dividing was exactly the problem it was built to solve.

So the clock on your wall, the protractor in a drawer somewhere, or the GPS spitting out coordinates in degrees are running on the exact same logic that was once a mere trick on someone's fingers, repurposed for warehouse inventory, aimed at the night sky. Nobody voted on it, nobody redesigned it, nobody replaced it since, because it happened to fit the two things humans never stopped needing to divide cleanly: the day and the circle.

How “Irrational” is Your Number? Chasing Irrationality Measures & Dancing Sequences: $(n\alpha)$ vs. $(p_n\alpha)$

1. The Ultimate Math Debate

It turns out irrationals have *personalities*. Some play nice and sit comfortably far from rational fractions, while others—the "rebel" Liouville numbers—sneak uncannily close to fractions p/q . To quantify this drama, we use the **irrationality measure** $\mu(\alpha)$.

The measure $\mu(\alpha)$ measures how closely rational fractions can shadow α :

$$\left| \alpha - \frac{p}{q} \right| < \frac{1}{q^\mu} \quad \text{for infinitely many } \frac{p}{q}$$

[illegible]

In the second half of my internship, I took these numbers and put them to work generating sequence multiples on the unit interval $[0, 1)$ by taking fractional parts $\{x\} = x - \lfloor x \rfloor$.

Round 1: $(n\alpha) \pmod{1}$

Take any irrational α and watch $\{\alpha\}, \{2\alpha\}, \{3\alpha\}, \dots$ spread across $[0, 1)$. Thanks to Weyl's Equidistribution Theorem, these points fill the interval *perfectly uniformly*!

By using *Weyl's Criterion*, we check the exponential sum:

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N e^{2\pi i h n \alpha} = 0 \quad (\forall h \neq 0)$$

Because this geometric series stays bounded while $N \rightarrow \infty$, uniform distribution is guaranteed!

Round 2: The Prime Challenge $(p_n \alpha) \pmod{1}$

What if we throw away regular integers and step only on *prime numbers* $p_n = 2, 3, 5, 7, 11, \dots$? Primes are chaotic and sparse—does order survive?

Vinogradov's Magic (1937)

Using deep exponential sum techniques, Vinogradov proved $(p_n \alpha)$ is **also equidistributed!** But during my numerical simulations in Python, I found huge differences in how fast they settle down:

- **Speed Limits:** Regular $(n\alpha)$ spreads rapidly with a low discrepancy $D_N \approx O(\frac{\log N}{N})$. Prime sequence $(p_n \alpha)$ converges much slower!
- **Clustering Traps:** When α has a high irrationality measure $\mu(\alpha)$, $(p_n \alpha)$ exhibits temporary "clumping" before finally smoothing out for huge N .

4. Internship Takeaways

This summer showed me that number theory isn't just static equations—it's a dynamic playground! Understanding how irrationality measures dictate sequence distributions is key to chaos theory, dynamic systems, and signal processing.

Got a favorite irrational number? Try coding a quick plot of $\{p_n \alpha\}$ in Python and watch the primes dance!

CHAPTER 3

Moheener Ghoraguli and the Mathematics of Musical Boundaries

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In 1970s Bengal, while the state was undergoing a period of political turmoil, a spark of change was simultaneously igniting Bengal's cultural landscape. Gautam Chattopadhyay with his fellows started a progressive music band *Moheener Ghoraguli*, arguably the first ever rock band across the country. Now you may wonder, what this has to do with Mathematics at all?

Hold your horses! (*Moheener Ghoraguli* literally means “Mohin's horses” and the phrase itself carries a literary echo of Jibanananda Das. Yes, that was a lame pun. The writer accepts full responsibility.)

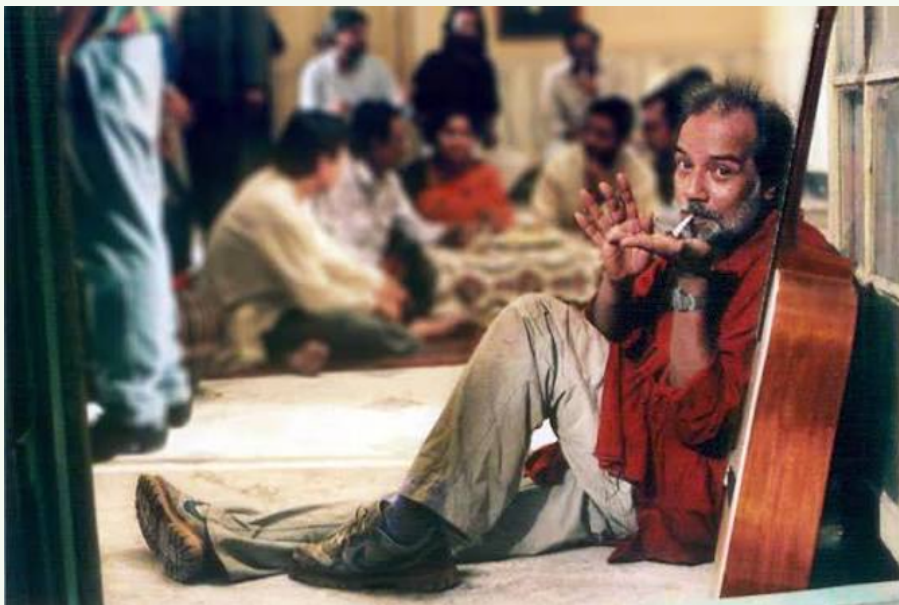


Figure 3.1: Gautam Chattopadhyay, founder of Moheener Ghoraguli.

Imagine you are on a family trip. Your sister insisted you play rock, but your father had a different taste, and hence, he recommended you play Bengali folk music.

You started wondering: there might be some features in music that distinguish one genre from another. But this boundary is not always clear at all. *Moheener Ghoraguli* started galloping in your head, and you realised that they took inspiration from Baul and American folk. But their music is also heavily influenced by blues rock. You could see the apparent contradiction in their musical styles and began wondering whether this blend of genres could be understood mathematically.

“*Dhadhar thekeo jotil tumi, khider thekeo sposto ...*”

While listening, you realised that a song is not that ‘jotil’ after all. An audio recording is essentially a signal whose amplitude varies with time. But we know that a broad class of functions can be represented as combinations of sine and cosine waves. Thus, we can decompose an audio signal into its constituent sinusoidal waves—a process known as Fourier decomposition.

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-i2\pi ft} dt \quad (3.1)$$

And with the Fourier transform of the song waveform, we could find what frequencies (musical notes, perhaps) that sound is made out of. $X(f)$ is indicating only that. But soon you will realise that we don’t really care about the total presence of any particular frequency in a song. Instead, we want to know at what time a particular frequency appears and its duration. Thus, the regular Fourier Transform won’t help us that much. Hence, instead of performing the Fourier transform over the entire song at once, we can take a small time window $w(t - \tau)$, isolate that short segment of the signal, and perform the Fourier transform on it.

$$X(\tau, f) = \int_{-\infty}^{\infty} x(t) w(t - \tau) e^{-i2\pi ft} dt \quad (3.2)$$

We then slide this window along the signal and repeat the process. Now $X(\tau, f)$ tells us not only which frequencies are present, but also how their presence changes with time. Voilà! you have just used the *Short-Time Fourier Transform* (STFT).

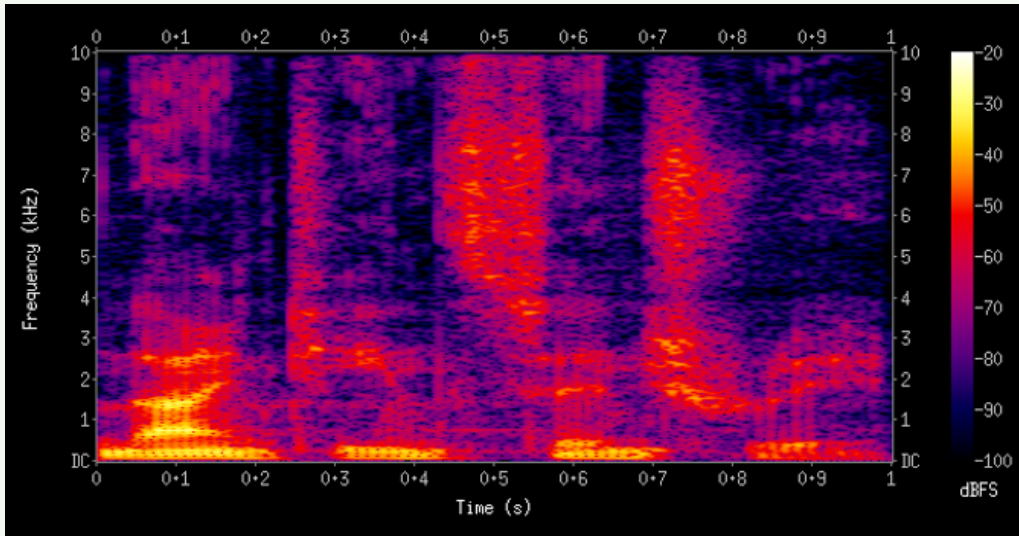


Figure 3.2: Spectrogram of a song.

The above plot is a *Spectrogram*, which is essentially a visual representation of the STFT. We could extract quantitative features about the song. Now, from a spectrogram at a given time, we could get a distribution of frequency and spectral energy around the frequency. With this only, we could produce a quantitative feature known as the *Spectral centroid*. Spectral centroid is analogous to the centre of mass, if we were talking about mass distribution. There is another feature called *Spectral flux*, which indicates how rapidly a spectrum is changing. There are more sophisticated features like MFCC’s—let’s

not worry about that. Now let's do a bit of abstraction. Let's consider our song as a vector (x) in a multidimensional feature space.

$$x = \begin{bmatrix} \text{Spectral Centroid} \\ \text{Spectral Flux} \\ \vdots \end{bmatrix} \quad (3.3)$$

The song vector is made out of all these quantitative features. Now you start to realise that our song is now an actual point in our multidimensional space. And every other song can be represented in exactly the same way. Each song therefore occupies its own point in this feature space, with its position determined by its individual musical characteristics. Thus, we will observe songs with similar genres clustering and forming some sort of cloud. Like, you may observe a Baul cloud and a Jazz cloud, and obviously, they will overlap sometimes.

Now consider the mean of all song vectors (N vectors) in a cloud.

$$\mu = \frac{1}{N} \sum_{i=1}^N x_i \quad (3.4)$$

That's our central feature of that cloud. So, naturally, we would like to know how far a particular song lies from the centre of its cloud. And how do we measure the distance between two points in a multidimensional space? Okay, okay—you already know the answer. Our old friend, Euclidean distance!

$$D_E(x, \mu) = \sqrt{\tau_E} \quad (3.5)$$

Euclidean distance has a problem—it treats each direction equally. The cloud may have large variation in one direction and less in another. Also, two features might be correlated. Thus, we need to take the covariance matrix into account, resulting in the *Mahalanobis distance* (named after P. C. Mahalanobis, founder of ISI and associated with the Physics Department, Presidency College).

$$D_M(x) = \sqrt{\tau_C} \quad (3.6)$$

Let's not get into the technicalities, but we can say that it measures the distance from the “statistical identity” of the cloud. Now, if you pick any song by “The Rolling Stones” (x_1), you will find maybe something like

$$D_M(x_1, \text{Rock}) = 0.3, \quad D_M(x_1, \text{Folk}) = 3.5 \quad (3.7)$$

But, on the other hand, if you pick a song by “Fakir Lalon Shah” (x_2), you will find maybe something like

$$D_M(x_2, \text{Rock}) = 5.8, \quad D_M(x_2, \text{Folk}) = 0.2 \quad (3.8)$$

Don't take these numbers at face value, but you get the idea. And when you try to do the same with most of Moheener Ghoraguli's songs, you observe that they don't solely belong to any particular genre cloud. And perhaps that is what makes Moheener Ghoraguli so interesting. Their music refuses to sit neatly inside one of our statistical clouds. Somewhere between Baul melodies, folk traditions, blues, jazz and rock, they occupy a region of their own. In the end, the mathematics does not tell us what a song

is. It merely gives us another way of looking at it. A melancholic baul melody from a Bengali village and a distorted guitar from Liverpool may sound worlds apart, yet in the right feature space, they may turn out to be neighbours.

Maybe the boundaries were never in the music to begin with.

CHAPTER 4

Can Geometry Emerge from Information?

From Graphs and Entropy to Spacetime

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“Space is not a thing. It is a relationship.”
 - paraphrased from Leibniz

1. A Graph, Nothing More

Consider the simplest mathematical object that encodes relationships: a graph

$$G = (V, E), \quad (4.1)$$

where V is a set of vertices and $E \subseteq \binom{V}{2}$ is a set of edges. No coordinates, no distances, no notion of “space”—only vertices and edges.

The question that animates this article is disarmingly simple:

Can the geometric properties we associate with physical space—distance, dimension, curvature—emerge purely from the combinatorial and information-theoretic properties of a graph?

This is not a metaphor. It is a precise mathematical programme, pursued actively under the banner of *causal set theory* [1], *quantum graphity* [3], and *emergent geometry* [4]. The starting point is pure discrete mathematics; the destination is the geometry of the universe.

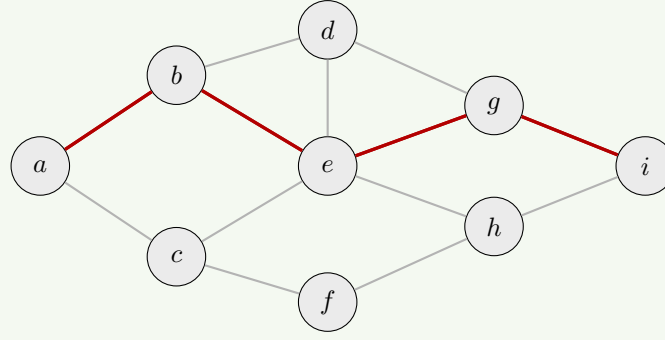
2. Distance Without a Ruler

On any connected graph $G = (V, E)$, there is a natural notion of distance. The *graph distance* $d_G(u, v)$ between two vertices $u, v \in V$ is the length of the shortest path connecting them:

$$d_G(u, v) = \min\{|P| : P \text{ is a path from } u \text{ to } v\}. \quad (4.2)$$

This function satisfies the axioms of a metric space: non-negativity, identity of indiscernibles ($d_G(u, v) = 0 \iff u = v$), symmetry, and the triangle inequality. Thus every connected graph is automatically a metric space (V, d_G) .

The remarkable observation is this: on certain families of graphs, the graph metric *approximates* a continuum metric. On the integer lattice $\mathbb{Z}^d$ with nearest-neighbour edges, d_G approximates the ℓ^1 distance on $\mathbb{R}^d$ at large scales. Geometry is not put in by hand—it is read off from combinatorics.



Shortest path: $d_G(a, i) = 4$

Figure 4.1: A graph G with nine vertices. The **highlighted path** from a to i has length 4, establishing the graph distance $d_G(a, i) = 4$. No coordinate system is needed—distance arises from connectivity alone.

3. Dimension from Growth

If distance can emerge from a graph, what about *dimension*? Define the *ball of radius r* around a vertex v :

$$B_r(v) = \{u \in V : d_G(u, v) \leq r\}. \quad (4.3)$$

In $\mathbb{R}^d$, the volume of a ball of radius r grows as r^d . We can turn this observation into a *definition*. The *growth dimension* (or Hausdorff dimension of the metric space) of a graph at vertex v is

$$d_{\text{growth}} = \lim_{r \rightarrow \infty} \frac{\log |B_r(v)|}{\log r}. \quad (4.4)$$

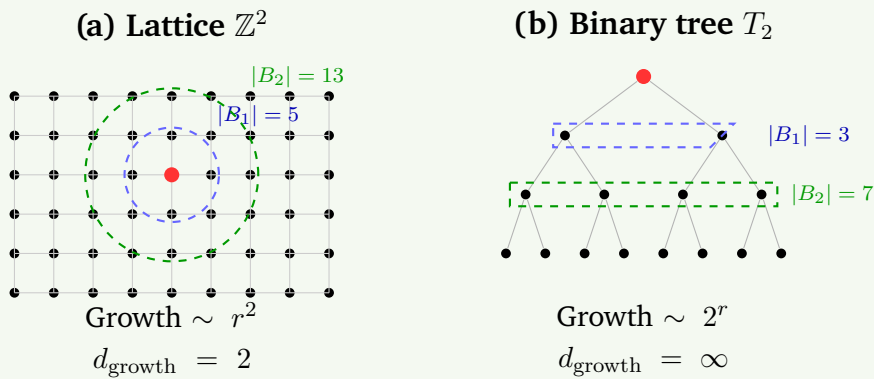


Figure 4.2: Ball growth determines dimension. **(a)** On the lattice $\mathbb{Z}^2$, $|B_r(v)| \sim r^2$, recovering dimension 2. **(b)** On a binary tree, $|B_r(v)| \sim 2^r$: the growth is exponential, reflecting the hyperbolic nature of trees. A graph “knows” its own dimension.

On $\mathbb{Z}^2$, we compute $|B_1| = 5$, $|B_2| = 13$, and in general $|B_r| \sim 2r^2 + 2r + 1$, so $d_{\text{growth}} = 2$ —the graph recovers the dimension of the plane. On a binary tree, $|B_r| = 2^{r+1} - 1$: the growth is exponential, reflecting the hyperbolic nature of trees. Thus, a graph *knows its own dimension*, without anyone declaring it.

4. Entropy: The Information Content of Structure

Shannon entropy quantifies information: given a probability distribution $\{p_i\}$,

$$H = - \sum_i p_i \log p_i. \quad (4.5)$$

But a graph also carries *structural* information. The *von Neumann entropy* of a graph G is defined via its normalised Laplacian. Let $L = D - A$ be the graph Laplacian, where D is the degree matrix and A is the adjacency matrix. Define $\tilde{L} = L / \text{tr}(L)$. Then:

$$S(G) = - \text{tr}(\tilde{L} \log \tilde{L}) = - \sum_{k=1}^n \tilde{\lambda}_k \log \tilde{\lambda}_k, \quad (4.6)$$

where $\tilde{\lambda}_k$ are the normalised Laplacian eigenvalues.

This entropy measures the “complexity” of the graph’s connectivity. A complete graph K_n (all-to-all connections) has *high regularity* but low informational content per edge. A graph with rich, heterogeneous structure carries more information.

The deep connection: in emergent-geometry proposals, the *entropy of the underlying discrete structure is conjectured to encode the area of surfaces in the emergent spacetime*. This is precisely the content of the Bekenstein–Hawking formula [2]:

$$S_{\text{BH}} = \frac{k_B A}{4 \ell_P^2}, \quad (4.7)$$

where A is the area of a black hole’s event horizon and ℓ_P is the Planck length. Information is geometry.

5. Causal Sets: When Order Becomes Spacetime

The most developed framework for emergent geometry is *causal set theory*, introduced by Bombelli, Lee, Meyer, and Sorkin [1]. The idea is radical yet simple:

Replace spacetime with a locally finite partially ordered set $(\mathcal{C}, \preceq)$.

A *causal set* (or *causet*) is a partially ordered set in which every *Alexandrov interval* $[x, y] = \{z \in \mathcal{C} : x \preceq z \preceq y\}$ is finite. The partial order represents *causal precedence*: $x \preceq y$ means “event x can influence event y .”

The *Hauptvermutung* (main conjecture) of causal set theory states:

If $(\mathcal{C}, \preceq)$ faithfully embeds into a spacetime manifold (M, g) , then the manifold, up to conformal equivalence, is uniquely determined by the order relation alone.

In other words, the causal order is the geometry. Dimension, topology, and metric—all are recoverable from $\preceq$. Partial results have been proved: the dimension d of the emergent spacetime can be estimated by counting the lengths of maximal chains (totally ordered subsets) in $\mathcal{C}$ [5, 6].

6. The Computational Perspective

For a computer scientist, this programme has a striking interpretation: spacetime is not a stage on which physics happens—it is the *output of a computation* performed on a discrete structure.

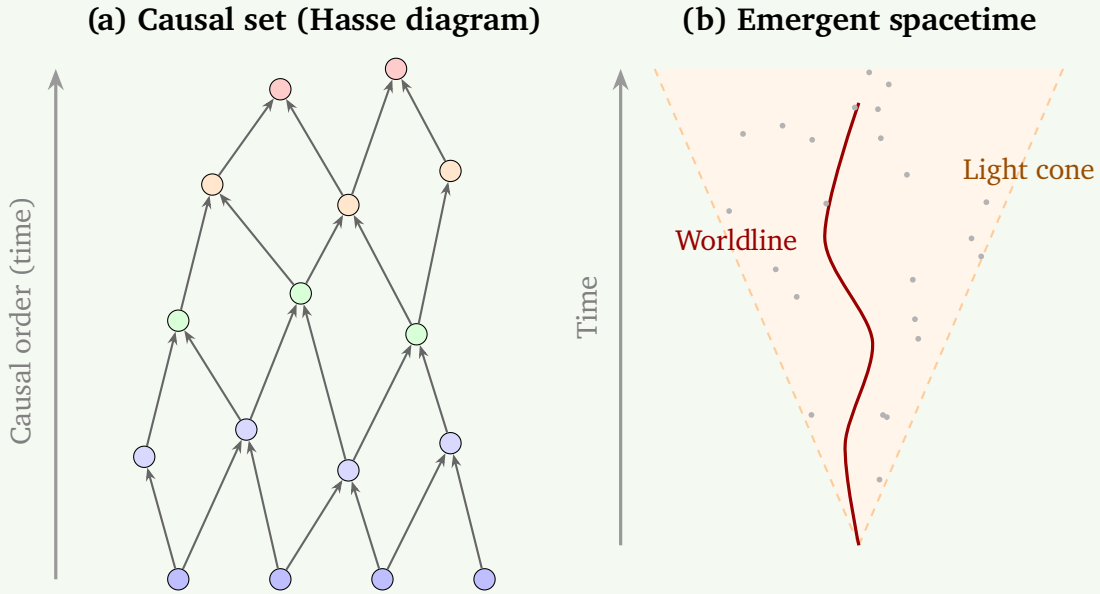


Figure 4.3: **(a)** A causal set: a finite poset whose elements represent spacetime events and whose order relation encodes causal precedence. **(b)** At large scales, the discrete causal structure is conjectured to approximate a smooth Lorentzian manifold, complete with light cones and worldlines.

Algorithms from computer science appear naturally:

- **Shortest-path algorithms** (Dijkstra, BFS) compute emergent distances.
- **Spectral methods** (eigenvalues of the graph Laplacian) determine dimension and curvature.
- **Entropy estimation** quantifies the information content of the structure.

The question “Can geometry emerge from information?” becomes, in the language of computation:

Is there an efficient algorithm that takes a sufficiently nice graph and outputs the geometric invariants of the space it approximates?

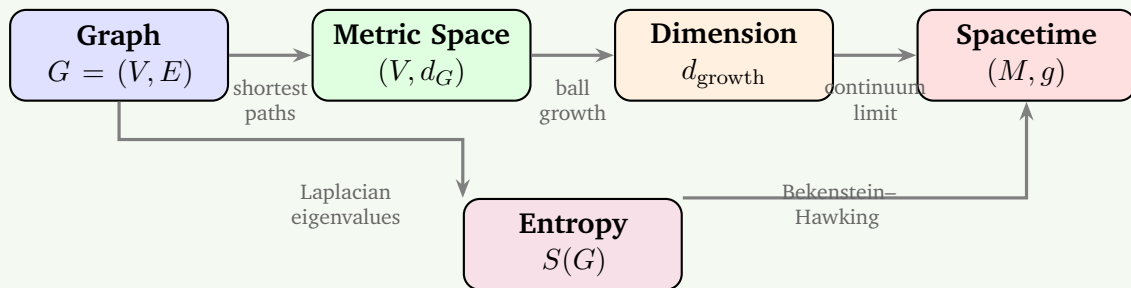


Figure 4.4: The emergence pipeline: from a graph G to spacetime (M, g) . Distance and dimension arise from combinatorics (top path). Entropy provides an independent route connecting graph structure to geometric area (bottom path). Both paths must agree for the programme to succeed.

7. Conclusion

We began with a graph $G = (V, E)$ - an object that a first-year student can draw on a napkin. From this object, using only combinatorics, analysis, and information theory, we extracted:

- **Distance** via the graph metric d_G ,
- **Dimension** via the growth rate of balls $|B_r(v)|$,
- **Information content** via the von Neumann entropy $S(G)$,
- **Causal structure** via partial orders $(\mathcal{C}, \preceq)$,
- and, tentatively, the **geometry of spacetime** itself.

The question is not whether the universe is a graph. The question is whether a discrete, relational structure-equipped with nothing more than order and counting-can give rise to the smooth geometry that general relativity describes.

This remains one of the most beautiful open problems at the intersection of mathematics, computer science, and physics. And it begins, as all good mathematics does, with a definition:

$$G = (V, E).$$

References

- [1] L. Bombelli, J. Lee, D. Meyer, and R. D. Sorkin, “Space-time as a causal set,” *Physical Review Letters*, vol. 59, no. 5, pp. 521–524, 1987.
- [2] J. D. Bekenstein, “Black holes and entropy,” *Physical Review D*, vol. 7, no. 8, pp. 2333–2346, 1973.
- [3] T. Konopka, F. Markopoulou, and L. Smolin, “Quantum graphity: a model of emergent locality,” *Physical Review D*, vol. 77, no. 10, 104029, 2008.
- [4] C. A. Trugenberger, “Combinatorial quantum gravity: geometry from random bits,” *Journal of High Energy Physics*, 2017(9), 045, 2017.
- [5] J. Myrheim, “Statistical geometry,” CERN preprint TH-2538, 1978.
- [6] D. A. Meyer, “The dimension of causal sets,” Ph.D. thesis, MIT, 1988.
- [7] R. D. Sorkin, “Causal sets: discrete gravity,” in *Lectures on Quantum Gravity*, A. Gomberoff and D. Marolf, eds., pp. 305–327, Springer, 2005.

CHAPTER 5

AI for Mathematics or Mathematics for AI?

A perspective for the next generation

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“The purpose of computation is insight, not numbers.” – Richard Hamming

The Two Visions

Have you ever stayed up late, staring at a single line of equations, wrestling with a proof until the quiet hours of the morning? That mix of frustration and exhilaration is the very pulse of mathematics. But something new is knocking on our classroom doors. Across the globe, AI systems are not just calculating—they are proving, conjecturing, and sometimes even surprising their human creators.

If you are an undergraduate mathematician today, you are living through a historical inflection point. The news is filled with terms like “Large Language Models,” “Machine Learning,” and “Formal Verification.” But what does this have to do with *you* and your studies?

Two opposing narratives dominate the discourse. The first suggests that we must develop increasingly sophisticated mathematics to fuel the next generation of Artificial Intelligence. The second suggests we should use AI as a tool to solve the deepest, most meaningful mathematical problems that have eluded us for centuries.

Honestly, it is tempting to pick a side and settle the debate. But this article is not about choosing one over the other. It is about understanding the landscape, recognizing the hidden landmines, and realizing that as a young mathematician, you are not merely a spectator in this revolution—you are the one holding the compass.

Mathematics for AI: building the digital brain

Let us burst a bubble right away. Artificial Intelligence is *not* magic. It is, at its core, a spectacular, large-scale application of the mathematics you are learning in your lecture halls right now.

Think about your Linear Algebra course. All those matrix multiplications, eigenvalues, and vector spaces? They are the native language of data. Every image, every word, every piece of information fed into an AI is first embedded into a high-dimensional vector space.

What about Calculus and Optimization? When an AI “learns,” it is performing gradient descent to minimize a loss function. You are taking derivatives of hundreds of billions of

variables simultaneously. And let us not forget Probability and Statistics, which provide the framework for dealing with the uncertainty of the real world.

If you choose to pursue this path, you are working on the *foundations of intelligence*. Every theorem you prove about convergence rates, every inequality you sharpen, directly translates to an AI that learns faster, generalizes better, and consumes less energy. In this paradigm, the mathematician is the architect. We build the rigorous infrastructure upon which the digital world rests.

For an undergraduate, this is an incredibly empowering thought: the abstract concepts in your textbooks are not just exercises for exams; they are the blueprints for tomorrow's technology.

AI for Mathematics: your personalized proof assistant

But imagine the opposite scenario. What if the student sitting in the lecture hall is not building the brain, but using that brain to prove the theorem?

In recent years, AI systems have begun to act as *co-pilots* for mathematicians. Tools like Lean, Coq, and Isabelle are proof assistants that can verify every logical step of an argument with inhuman precision. Imagine having a collaborator who never sleeps, never makes arithmetic mistakes, and can check the intricate details of a 200-page proof in seconds.

Consider the recent breakthroughs: AI systems have formalized and solved problems that have remained open for decades. They can generate conjectures by identifying patterns in data that the human eye might miss. They can test millions of potential counterexamples in a flash.

Picture this: you are working on a combinatorial number theory problem. You have a hunch, but the combinatorial explosion is too vast for manual checking. You feed the problem to an AI trained on thousands of mathematical papers. It suggests a lemma you never considered, one that fits your problem like a missing puzzle piece. It checks the boundary cases that would take you weeks.

For the undergraduate, this means the *method* of doing mathematics is changing. The lonely genius sitting in a room with a pencil and paper is now augmented by a non-human collaborator. The skill of the future is not just proving a theorem, but *formulating* the theorem in a way that an AI can assist with, and then *interpreting* the results the AI provides.

The Great Dilemma

So, where does the dilemma truly lie? You might be asking yourself, "Which side should I bet my career on?" Should we focus our intellectual firepower on improving the AI, or on using the AI to improve our mathematics?

The danger lies in treating these as mutually exclusive. If we focus solely on making AI smarter (Mathematics for AI), we risk turning our discipline into a service industry for tech conglomerates. We become the engine room for a machine we do not fully control.

On the other hand, if we focus solely on using AI to solve hard math (AI for Mathematics), we risk losing the *understanding*. A proof generated by a black box, even if verified by Lean, does not give us the *insight*—the "Aha!" moment that reconfigures our understanding of a whole field.

As Terence Tao, one of the greatest living mathematicians, puts it: AI can help with the *plumbing* of mathematics, but the *architecture* still belongs to humans. The thrill of

mathematics is not just in the destination of a correct proof, but in the journey of finding it.

The undergraduate must realize that the choice is not about which side to join, but about how to navigate the intersection. You must learn the rigorous math *and* the computational tools. You must know how to derive the gradient *and* how to query the model. But most importantly, you must never lose sight of *why* you are doing it.

What to do?

As your friendly neighbourhood mathematician, let me offer you some thoughts on navigating this new world:

1. **Master the fundamentals.** Do not skip the foundational courses, even if you have the option to do so. A strong grip on real analysis, abstract algebra, and logic is your armor. You cannot audit a black-box proof or design a robust algorithm without understanding the foundational building blocks.
2. **Learn to speak to the machine.** Pick up a programming language. You do not have to become a full-time software engineer, but you must be able to communicate with the machines. The worst outcome is a mathematician who is scared of the keyboard.
3. **Develop a moral compass.** Understand the history of technology and its social impact. Your mathematical training gives you a unique ability to distinguish truth from statistical noise and short-term successes. Do not underestimate the power of asking, “Should we build this?” rather than simply, “Can we build this?”
4. **Embrace the synthesis.** The future belongs to the *synthetic* thinker. The mathematician who can derive the backpropagation algorithm *and* question its epistemic limits. The mathematician who can ask an AI to search for a counterexample *and* explain the significance of the result to a room of fellow humans. Be that bridge.

The soul remains human

AlphaProof Nexus solved a 56-year-old problem, and that is magnificent. But it did so without joy. It solved it without the struggle, the despair, the late-night coffee, and the ultimate euphoria of the breakthrough.

The joy of discussing mathematics with fellow humans is far greater than any sterile interaction with an AI. Sharing a wild conjecture over tea, seeing a student’s eyes light up when they finally understand a tricky proof—these are the moments that make our field come alive.

AI will never know the aesthetic beauty of a symmetrical proof. AI will never carry the moral weight of deciding how its knowledge is deployed. AI will never inspire a room of anxious undergraduates to fall in love with the infinite.

You are the heir to a 4,000-year tradition of human inquiry. You are the ones who can build the mathematical engines of tomorrow *and* decide which destinations they navigate to.

Do not fear the dilemma. Embrace it. The mathematician of the future is not a builder or a user. The mathematician of the future is the one who ensures that while machines may calculate the answers, it is humanity that asks the right questions.

Further Reading:

- The Leiden Manifesto on Artificial Intelligence and Mathematics (2026).
- Terence Tao, “Mathematics in the age of AI” *Arxiv.org*.
- Hannah Fry, *Hello World: Being Human in the Age of Algorithms*.

CHAPTER 6

Mathematics and Literature: The Literary Side of Mathematicians

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A mathematician who is not also something of a poet will never be a complete mathematician.
-Karl Weierstrass

Mathematics is often perceived as a cold domain that stands rigidly on a vast framework of dry logic and hard rigor, which neither flows as freely as the verses of Milton nor possesses the moving quality of the prose of Jane Austen. Likewise, the masters of mathematics, who shaped the understanding of abstraction down the ages, seem to the unaware eye, to be serious and dull individuals who spend their lives scribbling meaningless symbols, which apparently have a lot of 'beautiful' information stored within them. However, these individuals, too, had a poetic side in them, which was also partly a quintessential element of their mathematical personalities.

To begin with, consider the first woman to earn a doctorate in Mathematics, Sofya Kovalevskaya. She worked extensively on Partial Differential Equations, extending and proving a landmark result by Cauchy. She also worked on the rotation of rigid bodies and the shape of Saturn's rings besides several other mathematical achievements. However, unexpectedly, she was also associated with Russian nihilism, feminism and political awakening of the late 19th century in Russia and elsewhere. In a way, she was a student of both Weierstrass and Dostoevsky, who was a regular guest to her parents' home when she was a child. As an author herself, she wrote several plays and produced two complete novels, *The Memories of Childhood* and *A Nihilist Girl*, with the former becoming a best-seller in Sweden. In her own words,

*The poet must see what others do not see, must see more deeply than other people.
And the mathematician must do the same.*

Her words echo the philosophy of her teacher, Karl Weierstrass, who believed that a mathematician should be a poet at heart having the artistic vision to see an elegant result or a pattern, or an abstract structure before rigor approves it.

The author of *Alice's Adventures in Wonderland*, Lewis Carroll, is remembered as the creator of nonsensical fantasies in the Victorian Era. Carroll is widely known for his wordplay, his satirical and amusing prose, creative poetry and short plays. Interestingly, Alice's creator was a mathematics lecturer at Christ Church, Oxford. His real name was Charles Dodgson and he worked in the fields of Linear Algebra, Geometry, Probability

and Formal Logic producing nearly a dozen mathematical textbooks. He wrote prolifically on inferences in logic, which he himself turned upside down in his novels. The Wonderland and the Mirror World of his works are parallel to the world of abstraction that a mathematician delights in; where initially nothing makes sense, but eventually everything is as clear as daylight!

The champion of Logicism, Bertrand Russell, changed the face of formal logic and the foundations of Mathematics forever through phenomenal works like the *Principia Mathematica* (co-authored with Alfred Whitehead). Yet again, he had a brilliant literary side that earned him his Nobel Prize in Literature in 1950. Curiously, his literary and mathematical works were published simultaneously and at times alternatively. Like Camus and Sartre, Russell produced countless books and essays on Philosophy, its problems, human morality, anti-war activism, religion and so on. His philosophical texts are as airtight as his proofs and both are remembered for their quality and impact. His philosophical angle of breaking down everything to the last atom is essentially the weapon he used so often in his mathematical works.

The discussion shall remain incomplete if Henri Poincaré's name is not mentioned. Widely known as *the Last Universalist*, Poincaré wrote popular essays on the philosophy of science, nature of science, human morality and some other humanistic questions he felt like exploring, besides his tremendous mathematical output. However his works are not merely acknowledged for their scientific or philosophical value, but also for the incredible quality of his prose and his deep thoughts for which he was elected to Académie Française in 1908, which represented the zenith of literary honour in France.

In his landmark work *La Science et l'Hypothèse* (1902), he described in his beautiful prose, the axiomatic setup and conventionalism in Geometry. His work soon became a literary sensation and influenced young Einstein, who would later use Geometry to model the Universe and even Picasso, who was moved when he heard these ideas from a friend who had read Poincaré's book. Picasso's early Cubist paintings can be seen as indirect artistic manifestations of Poincaré's ideas of Geometry. One genius could move the fields of Mathematics, Physics, Literature and even Art all at once!

Other mathematicians too had strong interests in literature, which also indirectly influenced their works. For example, Euler had a thirst for classical literature and ancient languages. Curiously, a verse describing a ship's motion in Book VI of Virgil's *Aeneid*, led him to contemplate the nature of motion of a vessel in water and in general the dynamics of fluids. One thought led to another and finally he could form the rigorous foundations of Analytical Fluid Mechanics. Gauss too had a serious interest in literature, rhetoric and the languages before his interests shifted towards mathematics and astronomy. Like the structured rhetoric he admired, he introduced systematic frameworks, new notations and a solid universal foundation for mathematicians to work upon. As a poet does, he obsessively revised his manuscripts, even destroyed ones which he felt mathematically inadequate and allowed only the best of his works to see the light of the world.

Like Gauss, Lagrange too might have been a Latin scholar had his interest not shifted to mathematics. He believed that a study of literature, grammar, and rhetoric is necessary for a developing mind to avoid burnout and to grasp the complex intellect and rigor

required in mathematics. He stressed the importance of literature throughout his life as it imparts the creative instincts and poetic vision an intellectual mind needs. Also he was the one to influence young Cauchy to master literature before turning to mathematics. Cauchy went a step further and produced some French and Latin verses later in life, which, however, were not as brilliant as his mathematics. For both of them, literature was not just a hobby, it was a training ground before serious mathematics, and a source of structured, systematic, creative and elegant understanding, which are necessary for mathematical thinking.

For these reasons and many more, it can be said that a mathematical mind is not intrinsically different from the mind of a poet, despite the massive differences of their spheres of work. A poet looks for beauty, rhythm, musical quality, deep themes and imagery while a mathematician looks for impeccable proofs, logic, rigorous frameworks to understand abstraction elegantly. In short, the poet and the mathematician both look for two things: truth and beauty (as Keats would have put it) and so it's no wonder that so many mathematicians in history were so fond of literature, which also shaped their own mathematical intuition from time to time, as the underlying creative impulse behind these apparently different fields is identical in spirit.

References

- [1] History of Science and Mathematics Stack Exchange. <https://share.google/FeAvvOLRk2XXoFcmn>
- [2] Bertrand Russell. https://en.wikipedia.org/wiki/Bertrand_Russell
- [3] Sofya Kovalevskaya. https://en.wikipedia.org/wiki/Sofya_Kovalevskaya
- [4] Lewis Carroll, *Through the Looking-Glass*, Puffin Classics.
- [5] Naresh Kinger (ed.), *World's Greatest Scientists*, Sahni Publications.
- [6] Lewis Carroll. https://en.wikipedia.org/wiki/Lewis_Carroll
- [7] Henri Poincaré. https://en.wikipedia.org/wiki/Henri_Poincar%C3%A9
- [8] Arthur I. Miller, *Einstein, Picasso: Space, Time, and the Beauty That Causes Havoc*.
- [9] ELOGIUM OF EULER. <https://share.google/Rah65uNsRHP0S6j19>
- [10] Augustin Louis Cauchy. <https://mathshistory.st-andrews.ac.uk/Biographies/Cauchy/>
- [11] Joseph-Louis Lagrange. https://en.wikipedia.org/wiki/Joseph-Louis_Lagrange
- [12] Joseph Louis Lagrange. https://www.maths.tcd.ie/pub/HistMath/People/Lagrange/RouseBall/RB_Lagrange.html
- [13] Ode on a Grecian Urn. *Poetry Foundation*. <https://www.poetryfoundation.org/poems/44477/ode-on-a-grecian-urn>
- [14] https://www.goodreads.com/author/quotes/1557580.Karl_Weierstrass

CHAPTER 7

The Building Blocks of Arithmetic: A Journey Through the Primes

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Introduction

Let's begin with an interesting question: Have you ever wondered why some numbers can be divided only by 1 and themselves, while others can be divided into smaller factors?

Numbers such as 2, 3, 5, 7 belong to a special group known as *prime numbers*. Now you may wonder why I haven't included 1 in the list.

The reason is that a prime number must have exactly two distinct positive divisors: 1 and itself. However, 1 has only one positive divisor: 1 itself. Therefore 1 is neither a prime number nor a composite number.

Prime numbers are considered the building blocks of natural numbers.

For example:

$$40 = 2 \times 2 \times 2 \times 5$$

The Fundamental Theorem of Arithmetic:

Every integer greater than 1 can be expressed as a product of primes in a unique way, apart from the order of the factors.

This raises an important question: *Are there infinitely many primes?*

Euclid's Proof: Are There Infinitely Many Primes?

Suppose, for the sake of contradiction, that there are only finitely many primes: $p_1, p_2, p_3, \dots, p_n$. Now construct the number:

$$N = p_1 \times p_2 \times \dots \times p_n + 1$$

Dividing N by any prime in the list leaves a remainder of 1, so none of them divides N . Since $N > 1$, it has a prime factor. This factor is not in our original list, contradicting our assumption.

For example, suppose we assume that the only prime numbers are 2, 3, 5, and 7. Their product is

$$2 \times 3 \times 5 \times 7 = 210.$$

Adding 1 gives us $N = 211$. None of 2, 3, 5, or 7 divides 211, so 211 is a new prime.

Thus, Euclid proved that there are infinitely many prime numbers.

Prime Gaps and Twin Primes: Exploring the Distribution of Primes

Although primes are infinite, they are not evenly distributed. The difference between consecutive primes is called a *prime gap*. For example, the gaps between 11 and 13 and between 23 and 29 are 2 and 6, respectively.

Let n be a positive integer. Consider the factorial:

$$(n + 1)! = 1 \times 2 \times 3 \times \cdots \times (n + 1)$$

It is divisible by every integer from 2 to $n + 1$. Consider:

$$(n + 1)! + 2, (n + 1)! + 3, \dots, (n + 1)! + (n + 1)$$

For every integer k from 2 to $n + 1$, the number $(n + 1)! + k$ is divisible by k and is greater than k . Therefore, each number is composite.

Thus, prime gaps can be arbitrarily large.

Conversely, *twin primes* are pairs of primes that differ by 2.

Examples include (3, 5), (5, 7), (11, 13), and (17, 19).

The **Twin Prime Conjecture** states that there are infinitely many pairs of twin primes. Although mathematicians have discovered many examples, a complete proof that infinitely many such pairs exist has not yet been found.

Fermat's Little Theorem

Fermat's Little Theorem states that if p is prime and p does not divide a , then

$$a^{p-1} \equiv 1 \pmod{p}.$$

For example, $2^4 = 16 \equiv 1 \pmod{5}$.

An equivalent form is $a^p \equiv a \pmod{p}$. It is also used in some methods of testing primality.

However, some composite numbers, called *pseudoprimes*, satisfy similar congruences. For example, $341 = 11 \times 31$ is composite, yet it satisfies $2^{340} \equiv 1 \pmod{341}$. Thus, Fermat's test may incorrectly identify some composite numbers as prime.

The Prime-Counting Function and the Prime Number Theorem

The prime-counting function, denoted by $\pi(x)$, is defined as

$$\pi(x) = |\{p \leq x : p \text{ is prime}\}|.$$

It represents the number of prime numbers less than or equal to x . The function behaves like a step function: it remains constant between consecutive primes and increases by 1 whenever x reaches a prime number. For example, $\pi(10) = 4$. When x increases from 10 to 11, the value becomes $\pi(11) = 5$ because 11 is prime. However, $\pi(12)$ remains 5 because 12 is not prime. At $x = 13$, the function increases again to $\pi(13) = 6$.

The **Prime Number Theorem** provides an approximation for $\pi(x)$ when x is large. It states that the proportion of prime numbers among the integers decreases as numbers become larger. Formally,

$$\lim_{x \rightarrow \infty} \frac{\pi(x)}{x / \ln(x)} = 1$$

Here, $\pi(x)$ represents the number of primes less than or equal to x . For example, $\pi(10) = 4$ because the primes not exceeding 10 are 2, 3, 5, and 7. Also, $\ln(x)$ denotes the natural logarithm of x . This result means that $\pi(x)$ becomes approximately equal to $x / \ln(x)$ as x tends to infinity.

The Reciprocal Series of Primes

Euler, the renowned Swiss mathematician, proved that the sum of the reciprocals of all prime numbers diverges:

$$\frac{1}{2} + \frac{1}{3} + \frac{1}{5} + \frac{1}{7} + \cdots = \infty$$

In other words, although prime numbers become increasingly sparse, the sum of their reciprocals continues to grow without bound. This result is surprising because it shows that primes are sufficiently abundant to produce an infinite sum, despite their irregular distribution.

Sieve of Eratosthenes: An Ancient Method of Finding Primes

How can we find all the prime numbers up to a given number without testing every number individually?

The **Sieve of Eratosthenes** is an ancient method developed by the Greek mathematician Eratosthenes to find all primes up to a specified number.

Suppose we want to find the primes up to 30:

Write down the integers from 2 to 30. Keep 2 and cross out all its multiples. Keep the next uncrossed number, 3, and cross out its multiples. Repeat the process with 5 and 7. The numbers that remain are prime.

The primes up to 30 are:

$$2, 3, 5, 7, 11, 13, 17, 19, 23, 29.$$

An important observation is that, to find primes up to a number N , we only need to consider prime numbers not exceeding $\sqrt{N}$.

Why? Every composite number has at least one prime factor less than or equal to its square root.

The Sieve of Eratosthenes shows how a simple, systematic process can reveal the hidden structure of numbers.

Conclusion

Prime numbers are simple to define but extraordinarily rich in structure. Euclid's proof establishes their infinitude, while prime gaps reveal the irregularity of their distribution. Twin primes present an unsolved challenge, and Fermat's Little Theorem shows their role in modular arithmetic.

The Prime Number Theorem describes their large-scale density, whereas Euler's result on reciprocal primes reveals their unexpected collective abundance. From the Sieve of Eratosthenes to modern sieve methods, the study of primes demonstrates how elementary questions can lead to profound mathematics.

Prime numbers therefore remain not only the building blocks of arithmetic, but also an enduring gateway to the exploration of infinity.

CHAPTER 8

Can we talk about irreducible and prime elements of general commutative rings?

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1. Introduction and Preliminaries

In most textbooks, prime and irreducible elements are defined only when the ambient ring is an integral domain. Have you ever wondered why this restriction is imposed? Or, put differently: what difficulties arise if one tries to define these notions over a general commutative ring, and can those difficulties be overcome? As we investigate this question, we first fix notation and record some well-known facts, with proofs, so that this note is reasonably self-contained for an undergraduate reader.

Throughout this note, *ring* will always mean a commutative ring with identity. In general, R will denote a commutative ring with 1. An element $u \in R$ is called a **unit** if $uv = 1$ for some $v \in R$. The set of units of R forms a group and is denoted by R^* . For $a, b \in R$, the elements a and b are said to be **associates** if $a = \lambda b$ for some $\lambda \in R^*$. For $a_1, a_2, \dots, a_n \in R$, the ideal generated by $a_1, \dots, a_n$ in R is denoted by $(a_1, \dots, a_n)$ or $a_1R + a_2R + \dots + a_nR$. An ideal I of R is called an **improper** ideal of R if $I = R$; otherwise I is called a **proper** ideal. For any two ideals I and J of R , the notation $I + J$ denotes the ideal generated by $I \cup J$ in R , and one checks easily that $I + J = \{x + y \mid x \in I, y \in J\}$. The polynomial ring over R with indeterminate X will be denoted by $R[X]$.

We will also make constant use of the following terminology.

Definition 1. Let R be a ring and $a \in R$ a non-zero element. We call a a **non-zero-divisor (NZD)** if, for all $b \in R$, $ab = 0$ implies $b = 0$. We call a a **zero-divisor (ZD)** if a is not an NZD, i.e., if there is a non-zero $b \in R$ with $ab = 0$. Thus, every non-zero element of R is either an NZD or a ZD, and not both. In this convention, 0 is neither an NZD nor a ZD.

Let us record a few basic properties of elements of a ring and the ideals they generate.

Lemma 1. Let R be a ring and $a, b \in R$. Then $a \mid b$ if and only if $b \in aR$, i.e., $bR \subseteq aR$. Moreover, if $b = \lambda a$ for some $\lambda \in R^*$, then $aR = bR$.

Proof. Since $a \mid b$, we have $b = ar$ for some $r \in R$, i.e., $b \in aR$; and $b \in aR$, and hence $bR \subseteq aR$. For the last statement, if $b = \lambda a$ with $\lambda \in R^*$, then $b \in aR$, so $bR \subseteq aR$ by the first part. Conversely, $a = \lambda^{-1}b \in bR$, so $aR \subseteq bR$. Together, $aR = bR$. $\square$

Lemma 2. Let R be an integral domain and $a, b \in R$. Then a and b are associates if and only if $a \mid b$ and $b \mid a$ in R , if and only if $aR = bR$.

Proof. If a and b are associates, say $a = \lambda b$ with $\lambda \in R^*$, then Lemma 1 (applied with the roles of a, b swapped) gives $aR = bR$, and in particular $a \mid b$ and $b \mid a$.

Conversely, suppose $a \mid b$ and $b \mid a$, i.e., $b = ax$ and $a = by$ for some $x, y \in R$. Then $a = by = (ax)y = a(xy)$, so $a(1 - xy) = 0$. If $a = 0$, then $b = ax = 0$ as well, and hence a and b are (trivially) associates. Now, suppose that $a \neq 0$. Then since R is an integral domain, $1 - xy = 0$, i.e., $xy = 1$; hence $x, y \in R^*$, and therefore, $a = by$ shows that a and b are associates.

Finally, $aR = bR$ is equivalent to $a \in bR$ and $b \in aR$ simultaneously (i.e., $bR \subseteq aR$ and $aR \subseteq bR$), which by Lemma 1 is the same as $a \mid b$ and $b \mid a$. $\square$

Lemma 3. Let R be a ring and I an ideal of R . Then, $I = R$ if and only if $\lambda \in I$ for some $\lambda \in R^*$; that is, I is improper precisely when it contains a unit.

Proof. If $I = R$, then certainly $1 \in I$, and $1 \in R^*$. Conversely, if I contains a unit λ , then since I is an ideal, $1 = \lambda^{-1}\lambda \in I$, and hence $r = r \cdot 1 \in I$ for every $r \in R$, i.e., $I = R$. $\square$

Lemma 4. Let R be a ring and $a, b \in R$ be such that $ab = 0$. If a is an NZD, then $b = 0$.

Proof. This is immediate from the definition of an NZD: a being an NZD means precisely that $ab = 0$ forces $b = 0$ for every $b \in R$. $\square$

We now recall prime and maximal ideals of a ring R . A **proper** ideal $I \subsetneq R$ of R is said to be **prime** if, whenever $a, b \in R$ satisfy $ab \in I$, then either $a \in I$ or $b \in I$. The ideal I is said to be a **maximal** ideal of R if I is proper and there is no proper ideal J of R with $I \subsetneq J$; equivalently, I is maximal if for every $x \in R \setminus I$ we have $I + xR = R$.

Proposition 1. Let R be a ring and I a proper ideal of R . Then the following hold:

1. I is a prime ideal of R if and only if R/I is an integral domain.
2. I is a maximal ideal of R if and only if R/I is a field.

Proof. Since I is proper, R/I is a non-zero ring; write $\bar{a}$ for the image of $a \in R$ in R/I .

(1) I is prime $\iff ab \in I \Rightarrow a \in I$ or $b \in I \iff \bar{a}\bar{b} = \bar{0} \Rightarrow \bar{a} = \bar{0}$ or $\bar{b} = \bar{0}$ in $R/I \iff R/I$ has no zero-divisors, i.e., R/I is an integral domain.

(2) Suppose I is maximal and $\bar{a} \neq \bar{0}$ in R/I , i.e., $a \notin I$. Since I is maximal, $I + aR = R$, so $1 = b + ar$ for some $b \in I, r \in R$. Going modulo I , i.e., in R/I we get $\bar{a}\bar{r} = \bar{1}$, so $\bar{a} \in (R/I)^*$. Hence R/I is a field.

Conversely, suppose R/I is a field. Let $a \in R \setminus I$; then in R/I we see that $\bar{a} \neq \bar{0}$, so $\bar{a}$ is invertible: $\bar{a}\bar{r} = \bar{1}$ for some $r \in R$. Thus $1 - ar \in I$, so $1 \in I + aR$, and therefore $I + aR = R$. This shows I is maximal. $\square$

Corollary 1. Every maximal ideal of a ring R is a prime ideal of R .

Proof. If I is maximal, then R/I is a field by Proposition 1(2). Since any field is an integral domain, it follows from Proposition 1(1) I is a prime ideal. $\square$

Let us recall the textbook definitions of prime and irreducible elements in an integral domain.

Definition 2. In an integral domain R , a non-zero, non-unit element x of R is called **irreducible** if, whenever $x = ab$ for some $a, b \in R$, either $a \in R^*$ or $b \in R^*$. We call x **reducible** if it is not irreducible, i.e., if $x = ab$ for some $a, b \in R \setminus R^*$ (read $R \setminus R^*$ as “not belonging to R^* ”).

Corollary 2. *Let R be an integral domain and $x \in R$ a non-zero, non-unit element. Then x is irreducible in R if and only if, whenever $x = ab$, then either $xR = aR$ or $xR = bR$.*

Proof. Suppose x is irreducible and $x = ab$. Then $a \in R^*$ or $b \in R^*$; if, say, $b \in R^*$, then $a = xb^{-1}$, so a and x are associates, and Lemma 2 gives $xR = aR$. The case $a \in R^*$ gives $xR = bR$ similarly.

Conversely, assume that $x = ab$ implies that $xR = aR$ or $xR = bR$. If $xR = aR$, then x and a are associates by Lemma 2, say $x = \lambda a$ with $\lambda \in R^*$; comparing with $x = ab$ and cancelling a (possible since R is a domain and $a \neq 0$, as $x = ab \neq 0$) gives $b = \lambda \in R^*$. The case $xR = bR$ similarly gives $a \in R^*$. So x is irreducible. $\square$

Definition 3. In an integral domain R , a non-zero, non-unit element x of R is called **prime** if, whenever $x \mid ab$ for some $a, b \in R$, then either $x \mid a$ or $x \mid b$; equivalently, whenever $ab \in xR$, then either $a \in xR$ or $b \in xR$.

Corollary 3. *Let R be an integral domain and $x \in R$ a non-zero, non-unit element. Then x is prime in R if and only if xR is a prime ideal of R .*

Proof. This is immediate: x prime $\iff ab \in xR \Rightarrow a \in xR$ or $b \in xR$, and hence directly from the definition of prime ideals it follows that xR is a prime ideal (note $xR \neq R$ since x is a non-unit, by Lemma 3). $\square$

Note 1. Observe that in a ring R , even though the zero ideal (0) satisfying “ $R/(0)$ is an integral domain” is equivalent to R itself being an integral domain (equivalently, (0) being a prime ideal of R), we never call 0 a prime element of R ! This is precisely why in Corollary 3 we excluded the case $x = 0$.

2. Prime and Irreducible Elements over General Rings

Irreducibility as minimality of an element, and maximality of the corresponding principal ideal

Over an integral domain, irreducibility captures the idea that an element cannot be broken into two non-trivial parts, where “trivial” means one of the two factors is a unit. From this viewpoint, a reducible element can be broken into “smaller” parts. To make this precise, we need an ordering that lets us compare elements as “smaller” or “bigger,” for which Lemma 1 above is exactly what we need.

Using that lemma, we define an ordering: for $a, b \in R$, we say b is **smaller** than a if $aR \subseteq bR$ – that is, if the ideal generated by b contains the ideal generated by a – equivalently, if $b \mid a$ in R . By Lemma 2, this relation is a partial order on the associate classes of an integral domain (antisymmetry – i.e., b smaller than a and a smaller than b forces $a = b$ up to associates – is exactly the content of that lemma). Now, if x is a reducible element, then there exists $a \in R$ that is “strictly smaller” than x : indeed, there are $a, b \in R \setminus R^*$ with $x = ab$, so $xR \subseteq aR$, and $xR \neq aR$ (else b would be a unit by Lemma 2), i.e., $xR \subsetneq aR$. Consequently, for $y \in R$, saying “ y is irreducible” should mean that y cannot be broken into “strictly smaller” elements – that is, “ y is a minimal element with respect to this ordering,” which is in turn equivalent to saying “the ideal yR is maximal among the *proper* principal ideals of R .” (This equivalence is exactly Corollary 2 above, read through the ordering.)

Extending the definition of irreducibility over general rings

If we wish to extend the notion of irreducible elements to general rings, we must decide whether the minimality (equivalently, the maximality of the corresponding principal ideal) discussed above can be preserved. Since this correspondence rests on Lemmas 2 and 1, and Lemma 1 already holds over any ring, we only need to check whether Lemma 2 extends to general rings.

Let us first see what goes wrong if we keep the integral-domain definitions of prime and irreducible elements verbatim over a general ring R . Suppose $a \mid b$ and $b \mid a$ (i.e., a is “smaller” than b and b is “smaller” than a), say $b = ad$ and $a = bc$ for some $c, d \in R$. Substituting the second equation into the first gives $b = (bc)d = bcd$, i.e., $b(1 - cd) = 0$. If b is not assumed to be an NZD, this does *not* force $cd = 1$, so we cannot conclude $c, d \in R^*$, and hence cannot conclude $aR = bR$. Since Lemma 2 is exactly what underlies the minimality picture of irreducibility, and we want to retain that picture, we are forced to assume b is an NZD; by symmetry, a must be an NZD as well. This gives:

Lemma 5. *Let R be a ring and let $a, b \in R$ be NZDs. Then a and b are associates if and only if $a \mid b$ and $b \mid a$ in R , if and only if $aR = bR$.*

Proof. One direction follows from Lemma 2. For the other direction, we let $a \mid b$ and $b \mid a$. Then, $b = ad$, $a = bc$ for some $c, d \in R$. As computed above, $b(1 - cd) = 0$. Since b is an NZD, by Lemma 4 we get $1 - cd = 0$, i.e., $cd = 1$, so $c, d \in R^*$. Now, $a = bc$ with c a unit shows that a and b are associates. The equivalence with $aR = bR$ follows from Lemma 1. $\square$

Note 2. Lemmas 1 and 5 extend the partial order discussed above to the non-zero NZD elements of a general ring.

Now, suppose we keep the definition of irreducibility of elements over a general ring R the same as that of an integral domain, i.e., call $y \in R$ irreducible whenever $y = ab$ implies either $a \in R^*$ or $b \in R^*$. If, say, $b \in R^*$, then Lemma 1 gives $yR = aR$; so in either case $yR = aR$ or $yR = bR$. However, without assuming y is an NZD, we cannot run this argument in reverse: knowing $yR = aR$ or $yR = bR$ does not let us conclude $a \in R^*$ or $b \in R^*$. Recall that for the case of integral domains, this converse used Lemma 2; and we do not have its corresponding version for general rings if we include ZDs. Since our only available tool here is Lemma 5, we must assume that y is an NZD to preserve the equivalence with minimality. This leads to a modified definition of irreducibility for general rings: **let R be a ring and y a non-zero, non-unit NZD. We call y irreducible if $y = bc$ implies $yR = bR$ or $yR = cR$.**

This definition can be simplified further: Suppose $y = bc$ and neither $yR = bR$ nor $yR = cR$ holds. Since $b \mid y$ and $c \mid y$, by Lemma 1 we have $yR \subseteq bR$ and $yR \subseteq cR$; and by our assumption, both containments are strict, and hence by Lemma 1 we have $b \notin R^*$ and $c \notin R^*$. In other words, a factorization $y = bc$ can only fail the condition “ $yR = bR$ or $yR = cR$ ” when *both* factors are non-units – and conversely, whenever $b \in R \setminus R^*$ satisfies $yR \subseteq bR$ (i.e., b is a non-unit divisor of y), we may write $y = bc$ for some $c \in R$, and demanding that this never happens strictly is exactly the condition below defining irreducibility:

Definition 4. Let R be a ring and $y \in R$ a non-zero, non-unit NZD. We call y **irreducible** if, whenever there exists $b \in R \setminus R^*$ with $yR \subseteq bR$, it follows that $yR = bR$.

With this definition, the equivalence between minimality and irreducibility is preserved, and we obtain:

Proposition 2. *Let R be a ring and let $y \in R$ be a non-zero, non-unit NZD. Then y is irreducible in R if and only if y is a minimal element with respect to the division ordering, if and only if yR is maximal among the proper principal ideals of R .*

Proof. All three phrasings say the same thing, once unwound. By definition, y is irreducible iff no $b \in R \setminus R^*$ satisfies $yR \subsetneq bR$. By the ordering introduced above (“ b smaller than y ” means $yR \subseteq bR$), this says exactly that no non-unit b is strictly smaller than y , i.e., y is minimal for the division ordering. Moreover, since bR ranges precisely over the proper principal ideals of R as b ranges over $R \setminus R^*$ (by Lemma 3, $bR = R$ exactly when $b \in R^*$), the same condition says yR is not properly contained in any proper principal ideal, i.e., yR is maximal among proper principal ideals of R . $\square$

We shall see through the examples in Section ?? that, even though our definition of irreducibility excludes ZDs, a zero-divisor can still behave like an irreducible element in the ideal-theoretic sense described above.

Extending the definition of prime elements over general rings

Next, let us extend the definition of prime elements to general rings. We saw above that, over an integral domain, a non-zero element is prime if and only if the corresponding principal ideal is a prime ideal (Corollary 3). A further important property of prime elements in an integral domain is that they are irreducible.

Proposition 3. *Let R be an integral domain and let $a \in R$ be a prime element of R . Then a is irreducible in R .*

Proof. If possible, let a be reducible, i.e., $a = bc$ for some $b, c \in R \setminus R^*$. Then $a \mid bc$ (trivially, since $bc = a$), and since a is prime, either $a \mid b$ or $a \mid c$. Say $a \mid b$, and write $b = ad$. Substituting into $a = bc$ gives $a = adc$, i.e., $a(1 - dc) = 0$. Since R is an integral domain and $a \neq 0$, we have $dc = 1$, i.e., $c, d \in R^*$ – contradicting $c \notin R^*$. This completes the proof. $\square$

Note 3. If R is a UFD, the converse also holds: irreducible elements are prime, and this is a defining characteristic of UFDs.

Thus, over an integral domain: **(1)** a non-zero element being prime is synonymous with the corresponding principal ideal being a prime ideal, and **(2)** prime elements are irreducible. We want a definition of prime elements over general rings that preserves both properties.

Suppose we keep the definition of prime elements in general rings identical to that in integral domains. One can easily check, exactly as in Corollary 3, that an element $a \in R$ is prime if and only if aR is a prime ideal – this part transfers without change, since the argument there never used that R is a domain.

Next, we check whether primes remain irreducible: the proof of Proposition 3 breaks down unless we assume a is an NZD (see Lemma 4, which is needed to cancel a from $a(1 - dc) = 0$). Hence, to retain “prime implies irreducible,” we must exclude ZDs, giving:

Definition 5. In a ring R , a non-zero, non-unit NZD element x of R is called **prime** if, whenever $x \mid ab$ for some $a, b \in R$, then either $x \mid a$ or $x \mid b$; equivalently, whenever $ab \in xR$, then either $a \in xR$ or $b \in xR$; or, equivalently, the ideal xR is a prime ideal of R .

Proposition 4. Let R be a ring and let $a \in R$ be a prime element of R (in particular, a is an NZD). Then a is irreducible in R .

Proof. Word for word the proof of Proposition 3 goes through: if $a = bc$ with $b, c \in R \setminus R^*$, primeness gives (say) $a \mid b$, so $b = ad$ and $a = adc$, i.e., $a(1 - dc) = 0$; since a is now assumed to be an NZD, Lemma 4 gives $dc = 1$, so $c \in R^*$, a contradiction. $\square$

In Section ?? we shall see that there are examples of ZDs in rings which behave like prime elements in the ideal-theoretic sense described above, even though our definition of prime element excludes ZDs.

3. Examples

We now illustrate the discussion above with two examples, using $(\bar{a})$ to denote the principal ideal generated by $\bar{a}$ in the relevant quotient ring $\mathbb{Z}_n$.

Example 1. Consider $R = \mathbb{Z}_4$. Since $\gcd(1, 4) = \gcd(3, 4) = 1$, we have $R^* = \{\bar{1}, \bar{3}\}$, and $\bar{2}^2 = \bar{0}$. So every non-zero element of $\mathbb{Z}_4$ is either a unit or a zero-divisor – there is no non-zero, non-unit NZD in R at all. In particular, our (NZD-based) definitions of prime and irreducible are vacuous here: no element of $\mathbb{Z}_4$ even qualifies as a candidate.

Nevertheless, $\mathbb{Z}_4/(\bar{2}) \cong \mathbb{Z}_2$ is a field, so $(\bar{2})$ is a maximal ideal of $\mathbb{Z}_4$, and hence certainly maximal among the proper principal ideals of $\mathbb{Z}_4$. One also checks directly that there are no $\bar{x}, \bar{y} \in R \setminus R^*$ with $\bar{2} = \bar{x}\bar{y}$ (the only non-units are $\bar{0}$ and $\bar{2}$, and $\bar{0} \cdot \bar{0} = \bar{0}$, $\bar{0} \cdot \bar{2} = \bar{0}$, $\bar{2} \cdot \bar{2} = \bar{0}$, none of which equal $\bar{2}$). So even though $\bar{2}$ is a zero-divisor – and hence is excluded by definition from being called “irreducible” – the ideal it generates is still maximal among proper principal ideals of R : $\bar{2}$ behaves exactly like an irreducible element in the ideal-theoretic sense, despite failing the NZD hypothesis. Moreover, since $(\bar{2})$ is a maximal ideal of $\mathbb{Z}_4$, Corollary 1 tells us $(\bar{2})$ is also a *prime* ideal of $\mathbb{Z}_4$; so $\bar{2}$ behaves like a prime element in the ideal-theoretic sense as well, again despite being a zero-divisor.

Example 2. Consider $R = \mathbb{Z}_6$. As above, $R^* = \{\bar{1}, \bar{5}\}$, and $\bar{2}, \bar{3}, \bar{4}$ are zero-divisors. One computes $\bar{2} = \bar{2} \cdot \bar{4}$, $\bar{3} = \bar{3}^2$, and $\bar{4} = \bar{4}^2$, so each of $\bar{2}, \bar{3}, \bar{4}$ factors as a product of two non-units – i.e., each is “reducible” in the naive multiplicative sense.

On the other hand, $\mathbb{Z}_6/(\bar{2}) \cong \mathbb{Z}_2$ and $\mathbb{Z}_6/(\bar{3}) \cong \mathbb{Z}_3$ are both fields, so $(\bar{2})$ and $(\bar{3})$ are maximal ideals of $\mathbb{Z}_6$, hence maximal among its proper principal ideals. Note that $(\bar{4}) = (\bar{2})$: indeed $\bar{4} = \bar{5} \cdot \bar{2}$ with $\bar{5} \in R^*$, so $\bar{2}$ and $\bar{4}$ are associates by Lemma 1, and they generate the very same ideal – in view of Lemma 5. So there are only two maximal principal ideals arising this way, $(\bar{2}) = (\bar{4})$ and $(\bar{3})$, generated by three zero-divisors. Just like in the previous example, all of $\bar{2}, \bar{3}, \bar{4}$ are zero-divisors and so are excluded from our definition of “irreducible,” yet each generates a principal ideal that is maximal among proper principal ideals of R – again illustrating that zero-divisors can behave like irreducible elements even though they fall outside the scope of our definition. Further, since maximal ideals are prime ideals (Corollary 1), the ideal $(\bar{2}) = (\bar{4})$ is a prime ideal, and hence $\bar{2}$ and $\bar{4}$ behave like prime elements of $\mathbb{Z}_6$, even though neither is an NZD.

4. Further reading and questions

Readers interested in a systematic treatment of irreducible and prime elements and, more generally, of factorization theory in rings with zero-divisors may consult Anderson and Chun [1]. There, the authors work with essentially the same ideal-theoretic definition of irreducibility used in Section 2 above, but *without* excluding zero-divisors from the outset; instead, they prove directly that if a is irreducible (meaning: a is a non-unit and $a = bc$ implies $(a) = (b)$ or $(a) = (c)$) and $(a) \subseteq (b) \subsetneq R$, then a must be a zero-divisor and b a non-zero-divisor. From this, they deduce that a is irreducible if and only if (a) is maximal among the proper principal ideals of R , or (a) is maximal among the principal ideals generated by zero-divisors of R . This is precisely the phenomenon we saw in Examples 1 and 2: the zero-divisors $\bar{2} \in \mathbb{Z}_4$ and $\bar{2}, \bar{3}, \bar{4} \in \mathbb{Z}_6$ each generate a principal ideal maximal among the principal ideals generated by zero-divisors of the respective ring, which is exactly what makes them behave like irreducible elements.

Note that, in the examples of Section 3, every element of the rings involved is either a zero-divisor, a unit, or zero. However, this is a very special feature of small rings such as $\mathbb{Z}_4$ and $\mathbb{Z}_6$. For instance, $\mathbb{Z}_4[X]$ has many elements that are neither zero-divisors nor units – for example, $1 + X$ and $2 + X^2$. Neither polynomial is annihilated by a non-zero constant $c \in \mathbb{Z}_4$, so by Problem 1(5) below, neither is a zero-divisor in $\mathbb{Z}_4[X]$. Moreover, neither has a unit constant term with all remaining coefficients nilpotent; hence, by Problem 1(6), neither is a unit. To gain a better understanding of such rings, we invite the reader to solve the following problems.

Below, we give a set of problems for further study of the corresponding topics. The problems use terminology and notation found in standard textbooks, such as [2], [4], and [5]. For the more advanced topics and harder problems, one may refer to [3].

Problem 1. *Let R be a ring. Before attempting the problems below, keep the following fundamental point in mind: the zero ideal (0) is a prime ideal of R if and only if R is an integral domain, i.e., if and only if R has no zero-divisors. Thus, the failure of (0) to be prime is precisely what makes the theory of prime and irreducible elements subtle over a general ring, and each of the following problems is ultimately a manifestation of this failure. As a preliminary exercise, verify this equivalence (use Proposition 1(1) with $I = (0)$, and note that $R/(0) \cong R$). **Problems 4.1(2) and 4.1(4) are more challenging than the rest; the remaining problems are routine or moderate.***

1. Let $x, y \in R$ be non-zero elements with $xy = 0$. Show that there exist prime ideals P and Q of R such that $x \in P$ and $y \in Q$. [Hint: show that both x and y are non-units, and use Krull's theorem (every proper ideal is contained in a maximal ideal – you may take this as known), together with Corollary 1.]
2. (Assume in addition that R is Noetherian, and let $y \in R$ be a zero-divisor.) Show that y lies in a prime ideal of R that is itself an annihilator of a non-zero element. [Hint: choose $0 \neq z \in R$ with $yz = 0$, and let $P = \text{Ann}(w)$ be maximal (with respect to inclusion) among the ideals of the form $\text{Ann}(w)$ with $0 \neq w \in R$ and $wy = 0$; such a maximal P exists by the ascending chain condition. Show that P is prime: if $ab \in P$ with $b \notin P$, then $bz \neq 0$ and $\text{Ann}(bz) \supseteq \text{Ann}(z) = P$, so by maximality $\text{Ann}(bz) = P$, hence $a \in P$. Conclude that $y \in P$.]

3. Show that if every prime ideal of R is a maximal ideal of R , then every non-zero, non-unit element of R is a zero-divisor. [Hint: first show that every element of a minimal prime ideal of R is a zero-divisor, by localizing at that prime; then note that under the stated hypothesis every prime ideal is automatically minimal.] (For a converse under the additional hypothesis that R is Noetherian, see any standard text on associated primes.)
4. A zero-divisor x of R is nilpotent if and only if it belongs to every prime ideal of R . [Hint: this is a special case of the general fact that the nilradical of R is the intersection of all the prime ideals of R ; you may take this general fact as known, or prove it directly.]
5. $f \in R[X]$ is a zero-divisor of $R[X]$ if and only if there exists a non-zero $a \in R$ such that $af = 0$. (This is McCoy's Theorem; the "if" direction is immediate, and the "only if" direction may be proved by induction on $\deg f$.) Warning: it is tempting to guess that this is also equivalent to "every coefficient of f is a zero-divisor of R ," but this weaker-looking condition is not sufficient in general – find a ring R and a polynomial $f \in R[X]$ of degree 1, both of whose coefficients are zero-divisors of R , such that f itself is not a zero-divisor of $R[X]$. [Hint: try $R = \mathbb{Z}_2 \times \mathbb{Z}_2$.]
6. $f \in R[X]$ is a unit of $R[X]$ if and only if the constant term of f is a unit of R and all other coefficients of f are nilpotent.
7. If $e \in R$ is idempotent (i.e., $e^2 = e$) and $e \neq 0, 1$, show that e is a zero-divisor of R . [Hint: consider $e(1 - e)$.] Check this against the elements $\bar{3}, \bar{4} \in \mathbb{Z}_6$ from Example 2.
8. Let $n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$ with $p_1, \dots, p_k$ distinct primes and $a_i \geq 1$. Describe the units, the zero-divisors, and the nilpotent elements of $\mathbb{Z}_n$ in terms of the p_i and a_i . [Hint: use the Chinese Remainder Theorem, $\mathbb{Z}_n \cong \mathbb{Z}_{p_1^{a_1}} \times \cdots \times \mathbb{Z}_{p_k^{a_k}}$, together with the fact that a tuple in a product of rings is a unit (resp. a zero-divisor, resp. nilpotent) according to a simple condition on its components – work out what that condition is.]

Problem 2. Let R be a ring and $x \in R$ a non-unit NZD.

1. Show that if x is prime, then x is irreducible. (This is Proposition 4; write out the proof in your own words.)
2. Show that the converse fails in general: consider $R = \mathbb{Z}[\sqrt{-5}]$ and $x = 2$. Verify that 2 is irreducible in R , but not prime. [Hint: use the norm $N(a + b\sqrt{-5}) = a^2 + 5b^2$; if $2 = uv$, then $N(u)N(v) = 4$, and $N(u) = 2$ is impossible. For non-primality, note that $2 \mid 6 = (1 + \sqrt{-5})(1 - \sqrt{-5})$, but 2 divides neither factor in R .]

Problem 3. Let R be an integral domain. Show that the definitions of irreducible and prime elements given in this article (Definitions 4 and 5) coincide with the usual definitions recalled in Section 1.

References

- [1] D. D. Anderson and S. Chun, *Irreducible elements in commutative rings with zero-divisors*, Houston J. Math. **37** (2011), no. 3, 741–744.

- [2] D. S. Dummit and R. M. Foote, *Abstract Algebra*, 3rd ed., John Wiley & Sons, Hoboken, NJ, 2004.
- [3] M. F. Atiyah and I. G. Macdonald, *Introduction to Commutative Algebra*, Addison-Wesley, Reading, MA, 1969.
- [4] M. Reid, *Undergraduate Commutative Algebra*, Cambridge University Press, Cambridge, 1995.
- [5] R. Y. Sharp, *Steps in Commutative Algebra*, 2nd ed., Cambridge University Press, Cambridge, 2000.

CHAPTER 9

Where orthic triangles go

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The angle recursion

Let $T = ABC$ be nondegenerate, H its orthocentre, and $T' = A'B'C'$ the feet of the altitudes. Miquel's *pivot* theorem says that for points A', B', C' on the three sidelines the circles $(AB'C')$, $(BC'A')$, $(CA'B')$ concur. Here the pivot is H : the right angles at B' and C' put both on the circle with diameter AH , so $H \in (AB'C')$, and similarly at B and C (Figure 9.1). (The complete-quadrilateral form of Miquel's theorem is a different statement and does not apply.)

Those cyclic quadrilaterals give the recursion. If T is acute, $BC'HA'$ is cyclic on diameter BH , so $\angle HA'C' = \angle HBC' = \pi/2 - A$; likewise $\angle HA'B' = \pi/2 - A$, and adding gives $\angle B'A'C' = \pi - 2A$. If T is obtuse at A , then A, B, C, H is an orthocentric system, so T and the *acute* triangle BHC have the same altitude feet; applying the acute case to BHC , whose angles are $\pi - A$, $\pi/2 - C$, $\pi/2 - B$, gives $2A - \pi$ at A' and $2B, 2C$ at B', C' .

Theorem 1. With $\varepsilon = -1$ for acute T and $\varepsilon = +1$ for obtuse T , the labelled angles obey $\theta \mapsto \varepsilon 2\theta \pmod{\pi}$, and ε is exactly the ratio of the orientations of T' and T . Unsigned: acute $(A, B, C) \mapsto (\pi - 2A, \pi - 2B, \pi - 2C)$; obtuse at A , $(A, B, C) \mapsto (2A - \pi, 2B, 2C)$.

Ergodicity and Birkhoff

Normalise by π and let $\Sigma = \{(a, b, c) \in \mathbb{R}_{>0}^3 : a + b + c = 1\}$ with normalised Lebesgue measure μ . Theorem 1 gives $F: \Sigma \rightarrow \Sigma$, defined on $\Sigma_0 = \{a, b, c < \frac{1}{2}\}$ by $(1 - 2a, 1 - 2b, 1 - 2c)$ and on $\Sigma_A = \{a > \frac{1}{2}\}$ by $(2a - 1, 2b, 2c)$, cyclically.

Each Σ_i is a sub-simplex of measure $\frac{1}{4}$ and each $F|_{\Sigma_i}$ is an affine bijection onto Σ , expanding by 2 in every direction, with constant Jacobian 4. Hence μ is invariant: the

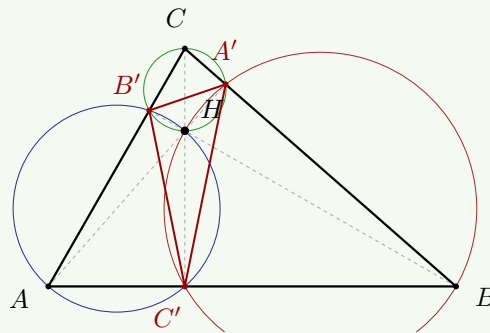


Figure 9.1: The three Miquel circles of the orthic configuration meet at H .

four inverse branches each pull Σ back to a quarter of itself with constant Jacobian $\frac{1}{4}$.

Theorem 2. F is ergodic with respect to μ , and $h_\mu(F) = \log 4$.

Proof. An n -cylinder is a set $C = \Sigma_{i_0} \cap F^{-1}\Sigma_{i_1} \cap \dots \cap F^{-(n-1)}\Sigma_{i_{n-1}}$. By induction F^n maps C affinely and bijectively onto Σ with constant Jacobian 4^n , so C is a simplex similar to Σ of diameter $2^{-n} \text{diam } \Sigma$, and $\mu(C) = 4^{-n}$.

Let E be invariant, $\mu(E) > 0$. Since $F^{-n}E = E$ we have $E \cap C = (F^n|_C)^{-1}(E)$, and constancy of the Jacobian gives

$$\frac{\mu(E \cap C)}{\mu(C)} = \mu(E) \quad \text{for every cylinder } C.$$

The cylinders containing a given point are simplices similar to Σ shrinking to it, so they form a regular family; at a Lebesgue density point x of E the left side tends to 1. Hence $\mu(E) = 1$.

The cylinders generate the Borel σ -algebra, so $\{\Sigma_0, \Sigma_A, \Sigma_B, \Sigma_C\}$ is a generating partition with $H(\cdot) = \log 4$ and all its refinements uniform; the coding map is a μ -a.e. bijection onto the one-sided full shift on four symbols carrying μ to the uniform Bernoulli measure, and Kolmogorov–Sinai gives $h_\mu(F) = \log 4$. $\square$

This recovers [3, 5, 1]; Manning’s tetrahedral shape space [4] is the unlabelled quotient. Birkhoff’s ergodic theorem now applies to any $\varphi \in L^1(\mu)$: for μ -a.e. starting shape, $\frac{1}{n} \sum_{k < n} \varphi(F^k x) \rightarrow \int \varphi d\mu$.

Corollary 1. Taking $\varphi = 1_{\Sigma_0}$: a.e. orbit is acute a quarter of the time, and each word of length k in the four types occurs with frequency 4^{-k} .

Area: the bound that fails

Euler’s pedal formula $[\text{pedal}(P)] = \frac{|R^2 - OP^2|}{4R^2} [ABC]$ with $P = H$ and $OH^2 = R^2(1 - 8 \cos A \cos B \cos C)$ gives

$$\frac{[T']}{[T]} = 2 |\cos A \cos B \cos C|. \quad (9.1)$$

Before the absolute value is taken, $\cos A \cos B \cos C$ is positive exactly on the acute region. So the power of the pivot with respect to the circumcircle fixes the symbol and the ratio at the same time.

The familiar bound $\cos A \cos B \cos C \leq \frac{1}{8}$, sharp at the equilateral shape, is a statement about the signed product. Off the acute region the product is negative, and the bound says nothing about its modulus there. Letting $A \rightarrow \pi$ with $B, C \rightarrow 0$ drives $\cos A$ to -1 and $\cos B, \cos C$ to 1, so the supremum of (9.1) is 2 and a single step can come close to doubling the area. Monte Carlo over 2×10^7 uniform shapes puts the measure of the expanding set at $\mu\{[T'] > [T]\} = 0.2261 \pm 0.0001$. The area sequence is not monotone, and no step-wise contraction estimate is available.

An exact contraction law

The decay rate of $[T_n]$ can be extracted from (9.1) by an ergodic average. There is no need.¹

¹We came to (9.1) first and spent some effort on $\int \varphi d\mu$ before noticing Lemma 1, which makes the integral unnecessary. It is kept below because the two computations have to agree, and that agreement is the only argument we have for $\int \log |\cos A| d\mu = -\log 2$ that is not a calculation.

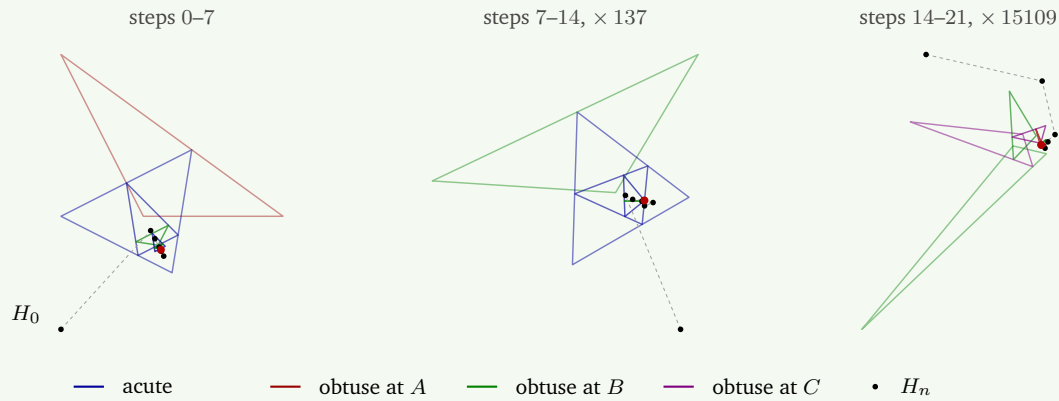


Figure 9.2: The same orbit in three windows, each magnified so that the triangles fill the frame. The orthocentre H_n generates T_{n+1} and is itself carried to ℓ (red) at rate 2^{-n} ; at $n = 40$ we find $|H_n - \ell| = 7.7 \times 10^{-13}$ against $2^{-40} = 9.1 \times 10^{-13}$. Rescaling by 2^n leaves a picture with no preferred scale, which is Lemma 1 and Theorem 2 seen together.

Lemma 1. $R(T') = \frac{1}{2}R(T)$ always.

Proof. The altitude feet lie on the nine-point circle of T , of radius $R(T)/2$, and three non-collinear points determine their circumcircle. $\square$

Theorem 3. $R_n = R_0 2^{-n}$, and for every n ,

$$[T_n] = [T_0] 4^{-n} \frac{\sin A_n \sin B_n \sin C_n}{\sin A_0 \sin B_0 \sin C_0}.$$

Proof. Iterate Lemma 1 in $[T] = 2R^2 \sin A \sin B \sin C$. $\square$

So with $\psi = \log(\sin A \sin B \sin C)$ and $\varphi = \log([T']/[T])$,

$$\varphi = -\log 4 + \psi \circ F - \psi. \quad (9.2)$$

Corollary 2. $\lambda := -\lim_n \frac{1}{n} \log \frac{[T_n]}{[T_0]} = \log 4$ for every orbit avoiding the degenerate set, and $\log[T_n] + n \log 4$ is bounded above and not below, so its fluctuation is not diffusive: the Green–Kubo variance of φ vanishes and the central limit theorem degenerates. Along a.e. orbit $\limsup 4^n [T_n]/[T_0] = (3\sqrt{3}/8)/\prod \sin A_0$ and $\liminf = 0$.

Scale is rigid, by Lemma 1, so every departure from monotonicity in Theorem 3 comes from the shape factor. That failure of monotonicity is not a blemish on the argument. It is what forces the argument to take the shape it does. A bound $[T_{n+1}] \leq c [T_n]$ with $c < 1$ would make convergence to a point immediate and would leave ergodic theory nothing to do; Section 9 shows no such bound exists, so $\log 4$ is reachable only as an average, through Birkhoff, or through the exact law above.

The expanding steps are also legible. Area grows in one step exactly when $\sin A \sin B \sin C$ grows by more than a factor of 4, and that happens only when a near-degenerate shape recovers sharply. So the measure 0.2261 is a statistic about recurrence away from $\partial\Sigma$, read off from areas, and the expanding steps are what make Corollary 2 nontrivial. Without them $4^n [T_n]$ would be non-increasing, and its $\limsup$ and $\liminf$ would agree.

Applying Birkhoff to (9.1) has to return the same constant. Under μ each angle has marginal density $2(\pi - u)/\pi^2$; substituting $u \mapsto \pi - u$ and using $\int_0^\pi \log |\cos u| du = -\pi \log 2$

gives $\int_0^\pi u \log |\cos u| du = -\frac{1}{2}\pi^2 \log 2$, so $\int \log |\cos A| d\mu = -\log 2$ and $\int \varphi d\mu = \log 2 - 3 \log 2 = -\log 4$. Numerically $\text{Var}_\mu(\varphi) = \pi^2/4$ and $\text{Var}_\mu(\log |\cos A|) = \pi^2/12$ to 20 digits, so the three $\log |\cos \cdot|$ are pairwise uncorrelated; we have no proof of this.

Entropy against contraction

The shape map expands by exactly 2 in each of two directions, and by Lemma 1 lengths contract by exactly 2. Neither rate involves an average, so $h_\mu(F) = \lambda = \log 4$ and $h/\lambda = 1$. For the map ℓ carrying an initial shape to the limit point of its pedal sequence, a Ledrappier–Young heuristic would predict a Hölder exponent $h/\lambda < 1$, the usual signature of a self-affine limit; with the ratio equal to 1 that prediction says nothing. The borderline case can be examined by counting terms. A perturbation of size ε separates the shape orbits at rate 2^n , and step n contributes $O(2^{-n})$ to the position of the limit point, so each of the first $\log_2(1/\varepsilon)$ steps contributes $O(\varepsilon)$ and the tail contributes $O(\varepsilon)$ in all.

Conjecture 1. $|\ell(x) - \ell(y)| \asymp |x - y| \log(1/|x - y|)$: the limit map is log-Lipschitz and of Zygmund class, not Lipschitz, and nowhere differentiable.

Manning proved that ℓ is continuous and not differentiable [4]. The count above says where the non-differentiability comes from. The difference quotient diverges logarithmically rather than as a fractional power, and it does so because the expansion and contraction rates coincide exactly. Difference quotients of ℓ along a line in Σ , computed at a sequence of scales, would settle it.

References

- [1] J. C. Alexander, *The symbolic dynamics of the sequence of pedal triangles*, Math. Mag. **66** (1993), 147–158.
- [2] J. G. Kingston and J. L. Synge, *The sequence of pedal triangles*, Amer. Math. Monthly **95** (1988), 609–620.
- [3] P. D. Lax, *The ergodic character of sequences of pedal triangles*, Amer. Math. Monthly **97** (1990), 377–381.
- [4] A. Manning, *The limit of a pedal sequence of triangles*, Bull. Lond. Math. Soc. **42** (2010), 673–679.
- [5] P. Ungar, *Mixing property of the pedal mapping*, Amer. Math. Monthly **97** (1990), 898–900.

CHAPTER 10

The Boy Who Modernized Mathematics

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Solving an unsolved problem for over 250 years is not everyone's cup of tea. While many mathematicians construct their theory brick by brick taking decades after decades, a young boy from Paris was like a thunder to the world of mathematics. He proved the famous result in equation theory at his teenage. A general polynomial equation of degree 5 or above can't be solved by radicals. The Quadratic Formula (Śrīdharācārya's formula) $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ is used to solve a general quadratic equation $ax^2 + bx + c = 0$, where $a, b, c \in \mathbb{R}, a \neq 0$. This is an example where we directly find the zeroes of the polynomial $ax^2 + bx + c$ in the field $\mathbb{R}$ by performing some basic Arithmetic operations on the coefficients like addition, subtraction, multiplication, division, exponentiation and square root operation. This is what exactly mean by the term "solving by radicals". Ofcourse there exist a formula for general cubic equations also but that is too dangerous to look at. Just search it on Google. However, he proved that by using radicals we can't directly find zeroes of a general polynomial of degree 5 or above which was an unsolved problem till then. His method of proving this result was so elegant that the tools he invented to approach this problem are now used in various branches of mathematics, cryptography or other fields. He was the first who used the term "Group" (in French "Groupe") to denote a special structure in sets with respect to some binary operation. Nowadays, the study of symmetry is totally based on it. In fact, in cryptography finite fields are very very important for digital security and any finite field is called *Galois field* in the honour of his work in modernization of Algebra.

Évariste Galois was born on 25 October 1811 to Nicholas-Gabriel Galois and Adélaïde-Marie in France. At the age of 14 he started to build serious interest in mathematics. His teacher Louis Paul Émile Richard identified his brilliance at Lycée Louis-le-Grand, a French Secondary School where he was admitted in October 1823. Galois wanted to study in École Polytechnique, the most prestigious institution for mathematics in France at that time. Unfortunately he failed to clear the entrance exam. It was truly a heartbreaking moment for him that we all can feel as a student. It is necessary to remember that failure didn't stopped him from doing mathematics. In that same year, he got admitted to École Normale (then known as l'École préparatoire), a far inferior institution for mathematical studies at that time. In the following year, Galois published his first research paper on simple continued fractions. After that he submitted his two papers on theory of equations to Academy of Sciences. Augustin-Louis Cauchy reviewed these papers, but refused to accept them for publication. Reasons are still unclear.

Galois made another attempt for his dream institution in 1829 but he failed again. If we were in his position may be it was tougher to stay consistent for mathematics to us, but he was furious. A brilliant and brave soul for every reason. Instead of giving up, he wrote his memoir on theory of equation and submitted several times but it didn't

published in his lifetime. He sent it to Joseph Fourier who was the secretary of Academy of Sciences at that time, but luck was not in favour of Galois. Unfortunately Fourier died soon after and the memoir was lost. Since it was 1830, not 2026 keeping copies of data was not so easy thing. However, after all these sad events Galois didn't lose his interest in mathematics.

May be magic happens, when we refuse to give up. In that year, he published three papers back to back. One of those was about finding root of an equation numerically, another one was about number theory, where finite field was introduced for the first time. The third one leads to *Galois Theory*, where Galois connected the solvability of an equation to the properties of its corresponding Galois Group. This is now like the heart of *modern abstract algebra*. Galois proved that a polynomial equation is solvable by radicals if and only if its Galois group is a *solvable group*. Infact, from his work the concept of group (which we are studying now) was established as a fundamental mathematical object and we all know the importance of groups today. The most important part of our story is that all these stuffs are invented by a 19 years old boy from Paris. It is important to mention the work of *Niels Henrik Abel* here, because he was the first who proved that a general quintic equation $ax^5 + bx^4 + cx^3 + dx^2 + ex + f = 0$, where $a, b, c, d, e, f \in \mathbb{R}, a \neq 0$ can't be solved by radicals. Galois generalised this result for any general polynomial equation of degree greater than equal to 5 with completely different approach. Abel was his contemporary mathematician who also died early at the age of 26 years. The major contributions of Galois were in Algebra, Analysis, Galois Theory and Continued fractions. He is considered widely as the intellectual father of Abstract Algebra for his revolutionary work.

Apart from mathematics Galois was actively involved in Politics and it is the misfortune of mathematics that the career of Galois was highly affected by it and infact he was arrested for his political movements. However, we will not discuss about his political career. Some people believe that mathematicians at that time had different political views than that of Galois and it can be a potential reason for his not-so-smooth academic career. Who knows! But it is important to understand that despite having all the odds, he was like an unstoppable spirit.

In 30th May 1832, Galois was shot to his abdomen during a duel. The reason for the duel is still unclear and the killer of Galois had lost to history. There is a theory about his mysterious death is a love triangle, another theory says that it is because of his involvement in active politics. We don't really know exactly what was happened to him and may be we can't know the true identity of his murderer. The history of people was written by people who were in power. We can't really do anything but remember Galois for the contributions he made to mathematics. We should learn to be consistent towards our interest from Galois, the boy who wrote to his younger brother Alfred before his death:

"Ne pleure pas, Alfred ! J'ai besoin de tout mon courage pour mourir à vingt ans !"
(Don't weep, Alfred! I need all my courage to die at twenty!)

CHAPTER 11

What's your “Braid in space” moment?

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In April 2026 during the Artemis II mission, astronaut Christina Koch shared with us a picture (one for the story books). It featured her, looking back at our little dot in space, Earth, from inside the Artemis II spaceship. But what made that picture different from every other, is her tiny braid floating in zero gravity. That picture moved something in me. And later when I went on the internet, I knew it moved something in a lot of people. Especially a lot of young girls. There was something so beautifully human, in watching a braid, something that's been part of every culture in showcasing not only the divine feminine but the ordinary. An ordinary woman, who, in her own frame of reference, is special. We know that women have been in space before. From Tereshkova to our own Kalpana Chawla. Then why did that picture suddenly feel so emotional? Why did that certain picture get women all over the globe talking, hoping and celebrating?

Without any hardcore research, just for the sake of this essay, I'm calling it, the “**braid in space**” effect. It's a representation of a very household and mundane thing that makes one feel more at home with an otherwise foreign concept. We have seen it before. In a certain book series when the writer decides to add a POC character, in a superhero movie when the female character doesn't die and wins the battle, in real life when you see your elder sister succeed in life against all odds. It's about getting reassurance against all hurdles.

Watching an astronaut with a tiny braid in space gave countless young girls the confidence to want to do it themselves, too. Some of them may change their mind halfway. But some of them might become the next faces of another mission in space.

What's important is the belief that got planted because of a simple, very simple photo. We all need something to believe in. And I think with the world in a state of collapse around us, that belief system needs to be stronger than anything right now. All my life I've thought that, the belief, that motivation needs to come from something big. An extraordinary grade, a gold medal, some fancy words. Then the “braid in space” had its effect on me too. And I understood, maybe it can be the most overlooked thing. And it can be personal.

I read a lot about a lot of female mathematicians while in my first year of BSc. One of the reasons was me being absolutely furious that I didn't know any theorem with a woman's name. There I read about Sophia Kovalevskaya and about Sophie Germain. But I kept searching. Are there more? The simple answer to this is: Yes. There are many more. More than we can name. More than we can count. Even the women whose names are not planted in theorems and history, there are more.

I read about Hypatia. And I was enraged that I did not know about her before. She was a mathematician, astronomer, and philosopher of Alexandria who lived in a time when it wasn't preferable for a woman to be that knowledgeable. She is the earliest

female mathematician of whose life and work reasonably detailed knowledge exists. She was the daughter of Theon, who himself played a very crucial part in the preservation of Euclid's *Elements*. She continued her father's work after his passing and herself is credited in commentaries in Apollonius's *Conics* and Diophantus's *Arithmetic*.

In her time, she was the world's leading mathematician and astronomer and also a renowned teacher in Alexandria. And as history has seen countless times, once a person challenges existing beliefs with scientific reasoning they become a threat to the persisting societal standards. And the worst that could happen, happened. Rumours of her being tied to witchcraft spread; she was seized by an enraged mob, dragged forcibly from her chariot and publicly lynched. She was dragged to Caesarium (a church), stripped naked and brutally tortured until she died. They burned her remains to ashes along with her works. But could they? One cannot burn ideas. Her ideas were preserved by her students and all who knew and admired her.

"Reserve your right to think, for even thinking wrongly is better than not thinking at all."

— Hypatia of Alexandria, translated.

Personally I like to read about different topics that interest me at times. However sometimes amidst all the pressure of exams and coursework I lose my spark. It's really easy to lose the spark but equally hard to get it back. 2 years into my BSc programme, heavily burnt out for different reasons, finding out Hypatia's story was sort of a "Braid in space" moment for me. This got me reading more about the women of mathematics who conquered not only intricate problems of mathematics and philosophy but also society. Again and again, those who broke all social barriers to practice their subject, to speak for it, do not get as much recognition as they deserve. Well, history has been pretty rough on women anyways. I think as a young girl just entering the field of mathematics, it would motivate 100 times more, had I known a little bit more about Maria Agnesi — who wrote the first comprehensive textbook on both differential and integral calculus; Mary Somerville — one of the first two women to be elected to the Royal Astronomical Society; Ada Lovelace — the first to write an algorithm designed to be run by a machine and Sofia Kovalevskaya — the first woman in modern Europe to gain a full professorship in mathematics.

I believe everyone should have their own "braid in space" moment. Simply that one story, one incident or one thing that weirdly gets them going. Maybe there's no active relation between the motivation to work and that incident, but then again, who knows how human brains work.

But while writing this essay I still kept thinking that this "braid in space" moment can be a little more personal. Why was I jarred by Hypatia's story so much? Then I remembered an old friend. She aspired to be a mathematics student, she was brilliant at it too. Sadly, she didn't end up pursuing mathematics, just like the stories of most middle class girls from a small town. I guess, if you're a girl from a small town, and you've aspired to be a student of physics, mathematics or engineering, you might have heard the famous punchline that goes "Pure sciences is for boys, girls should do nursing or medical" at least once in your life.

My dad taught me mathematics as a kid. He always said that the Mathematics field doesn't have enough women in it. And that's true. It has many. Many amazing women with amazing works. Still somehow it's not enough. I went to an all women's mathematics workshop. I can confirm from personal experience that I've never felt more understood. I

had a professor there whom I admired so much. She told me that she may be a mother, a wife and a sister; but before anything, she was a mathematician, a teacher, a professor.

These are all my "braid in space" moments. Watching my friend's dream shatter, my parents supporting me through it all, getting to learn from and about amazing female mathematicians and professors along the way. Something does stir inside my head when I read about the female scholars in history, their work being stolen and being severely underrepresented. But something also stirs inside my head when I see women succeed more and more in this field. In any field to be honest but this is the field I'm biased about.

*"Study mathematics. A very few girls do. Become good. Believe in yourself." —
My grandmother to me, a year before passing. Roughly translated.*

I think in a world that is becoming so relentless, it's very important to remember the very few things that keep us human. May they be stories of your favourite figures, may that be your art, may that be you sitting with a mathematics problem, stubborn enough not to use AI even if it takes all day. May we all have our "braid in space" moments that make us a little more human.

These were mine—
What's yours?

CHAPTER 12

The Mathematics of the Rat Race

Why Competition Can Make Caution Rationally Impossible

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Suppose that increasing effort is usually beneficial, but after a certain threshold it also creates a chance of a costly setback. If we are alone, risk aversion can make stopping early rational. But what happens when someone else is trying to beat us? A threshold-triggered stochastic payoff model gives a surprisingly sharp answer. The same risk that makes an individual cautious, can, in a competition, push both players toward maximum effort. This article develops the mathematics behind that paradox.

1. A simple question about effort

Imagine an athlete preparing for an important competition. Up to some point, more training reliably improves performance. However, beyond that point, training harder can increase the risk of injury or burnout. There is a natural question: If pushing harder becomes risky, when should a rational person stop?

The answer becomes much more interesting when there is an opponent.

Let $e \in [0, e_{\max}]$ denote effort, and let T be a threshold separating a “safe” region from a stochastic one. For $e \leq T$, suppose the payoff is simply

$$p(e) = e.$$

For $e > T$, define $x = e - T$. The player now faces a gamble:

$$p(e) = \begin{cases} e - S, & \text{with probability } q(e), \\ e + \beta(e - T), & \text{with probability } 1 - q(e), \end{cases}$$

where S is the severity of a shock and $\beta \geq 0$ measures the gain from successful extra effort.

The probability of a shock is assumed to increase with the distance beyond the threshold:

$$q(e) = 1 - \exp\left(-\frac{\alpha}{R}(e - T)\right), \quad R = e_{\max} - T,$$

where $\alpha > 0$ controls how rapidly risk accumulates.

Thus, crossing T does not simply make effort “bad.” It creates a genuine gamble: more effort produces both more upside and more downside.

2. The lone decision-maker: return versus risk

A risk-neutral person looks only at expected payoff. For $e > T$,

$$\mathbb{E}[p(e)] = e + (1 - q(e))\beta(e - T) - q(e)\bar{S},$$

where $\bar{S} = \mathbb{E}[S]$.

In the baseline model this expected payoff can continue increasing all the way to $e_{\max}$. So expected value alone does not produce caution.

This is where Markowitz's mean-variance framework becomes useful. Markowitz's 1952 paper on portfolio selection proposed thinking of expected return as desirable and variance as undesirable. In his words, the investor should consider "expected return a desirable thing and variance of return an undesirable thing" [1].

Adapting that idea to the present problem gives

$$U(e) = \mathbb{E}[p(e)] - \gamma \text{Var}[p(e)], \quad \gamma \geq 0.$$

The parameter γ measures risk aversion. A sufficiently large γ can make the increase in variance outweigh the increase in expected payoff. The optimum can then occur strictly inside the interval:

$$T < e^* < e_{\max}.$$

The important point is conceptual as well as mathematical. The model does not say that extra effort is necessarily irrational. It says that once uncertainty enters, "more expected payoff" and "better decision" are no longer synonymous.

The mean-variance expression is a standard approximation to expected utility under suitable conditions, but it also has a limitation. Variance does not distinguish between different forms of skewness. That limitation is particularly relevant here because the shock and gain branches are deliberately asymmetric [5].

3. A danger point without choosing a utility function

There is another way to ask when caution becomes justified. Instead of specifying a particular utility function, compare two adjacent choices.

At effort $e_k > T + 1$, consider the worst event at e_k : a shock occurs. Its probability is

$$q(e_k) = 1 - \exp\left[-\frac{\alpha}{R}(e_k - T)\right].$$

Now compare it with the best event at the previous effort $e_k - 1$: no shock. Its probability is

$$1 - q(e_k - 1) = \exp\left[-\frac{\alpha}{R}(e_k - T - 1)\right].$$

Define

$$f(e_k) = q(e_k) - [1 - q(e_k - 1)].$$

The "danger point" occurs when these probabilities are equal:

$$f(e_k) = 0.$$

Let

$$x = \frac{\alpha}{R}, \quad m = e_k - T.$$

Then

$$1 - e^{-xm} = e^{-x(m-1)}.$$

$$\text{Rearranging, } 1 = e^{-xm} + e^{-x(m-1)} = e^{-xm}(1 + e^x),$$

$$\text{so } e^{xm} = 1 + e^x.$$

$$\text{Therefore } m = \frac{1}{x} \ln(1 + e^x),$$

and hence the critical effort is

$$e_k^* = T + \frac{R}{\alpha} \ln(1 + e^{\alpha/R}).$$

This is a useful mathematical surprise. A particular utility function is not needed to identify the point at which the probability of the “worst next step” catches up with the probability of having avoided the shock one step earlier.

In the limiting case, the corresponding boundary condition is governed by the universal constant $\ln 2$. Thus the same model contains a clean analytical signature of when an internal danger point can exist [5].

4. Then an opponent enters

Now replace the lone decision-maker with two players. Both choose effort simultaneously, and whoever receives the larger realized payoff wins.

This changes the problem completely.

A Nash equilibrium, following John Nash’s framework for non-cooperative games, is a strategy profile from which no player can improve by changing their own strategy alone [2, 3]. Notice what this definition does *not* require: it does not ask whether the two players would be better off if they coordinated.

Suppose your opponent chooses a very high effort. If you respond by being cautious, you may reduce your own risk, but you also make yourself easier to beat. In the model, this strategic effect overwhelms the individual risk-return trade-off.

The resulting symmetric equilibrium is

$$(e_1^*, e_2^*) = (e_{\max}, e_{\max}).$$

In other words, both players escalate to the maximum feasible effort.

This is the central paradox of the model:

$$\boxed{\text{alone: risk can induce caution} \quad \longrightarrow \quad \text{competition: risk can induce escalation.}}$$

The competitive result is particularly striking because it does not require risk to disappear. The danger remains exactly where it was. What changes is the player’s objective, winning against another player replaces the individual optimization problem.

5. The rat race

This is closely related to George Akerlof’s idea of the “rat race”, in which individually rational competitive responses can produce collectively excessive outcomes [4]. The present model gives that intuition a specific stochastic mechanism.

Two students might each know that studying through the night is unhealthy. Two

athletes might each know that excessive training increases injury risk. Two investors might each recognize that increasing exposure is dangerous. Yet if backing down makes one lose relative to the other, neither player has an incentive to be the first to stop. The equilibrium is therefore not necessarily the outcome either person would choose in isolation. It is the outcome produced by mutual best responses.

There is also a mathematical clue about how to stop the escalation. If a certain, sufficiently convex cost of effort is added, for example

$$c(e) = \kappa e^2,$$

then the cost eventually grows fast enough to counteract the competitive benefit of pushing harder. An interior competitive equilibrium can then reappear.

6. One model, three answers

The same payoff structure therefore gives three different mathematical perspectives:

Perspective	Question	Typical conclusion
Mean–variance	“How much risk can I tolerate?”	interior optimum possible
Danger point	“When does one more step become dangerous?”	closed-form threshold
Game theory	“What if someone is trying to beat me?”	maximum effort equilibrium

That is perhaps the most interesting lesson. Rationality is not a single answer to a problem, it depends on the problem being solved.

When we are alone, we can compare reward with risk. When we compete, we must also compare ourselves with another decision-maker. And sometimes the presence of that opponent makes the cautious choice impossible to sustain.

References

- [1] H. Markowitz, *Portfolio Selection*, The Journal of Finance, 7(1), 77–91, 1952. <https://doi.org/10.1111/j.1540-6261.1952.tb01525.x>
- [2] J. F. Nash, *Equilibrium Points in N-Person Games*, Proceedings of the National Academy of Sciences, 36(1), 48–49, 1950.
- [3] J. F. Nash, *Non-Cooperative Games*, Annals of Mathematics, 54(2), 286–295, 1951.
- [4] G. A. Akerlof, *The Economics of Caste and of the Rat Race and Other Woeful Tales*, The Quarterly Journal of Economics, 90(4), 599–617, 1976.
- [5] K. Gupta, *Effort Under Threshold-Triggered Risk: A Stochastic Payoff Model and Its Game-Theoretic Implications*, 2026.

CHAPTER 13

The Topography of Silent Thoughts

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Imagine dropping a plastic chip onto a board studded with brass pegs. In a game of Plinko, the chip bounces haphazardly and rattles unpredictably down toward a dozen open slots at the bottom. A nudge of a fraction of a millimeter at the top can completely change which prize the chip lands in. Now, imagine a subtler game. The board still has pegs, but at the very bottom, there is only a single slot. Every chip, no matter where it drops, will eventually fall into that exact same destination. Yet, while some chips slide down almost instantaneously in a straight, gentle path, others get caught in complicated tangles, bouncing endlessly between pegs for a surprisingly long time before finally dropping through. This physical picture offers quite an accurate lens into what happens inside a new class of thinking machines when they encounter a difficult problem. When we interact with standard large language models, they generate output “out loud,” producing one visible token after another. A different paradigm, recurrent-depth reasoning models, thinks silently in an internal workspace before delivering an answer. Instead of stretching a model out across a massive sequence of feedforward layers, these architectures pass a problem through the same internal neural circuit repeatedly. Mathematically, the machine’s internal state at step t is represented by a high-dimensional

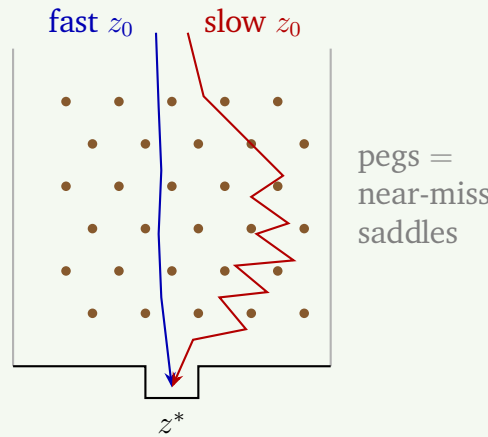


Figure 13.1: The single-slot board. Two nearby initial states z_0 both reach the same fixed point z^* , but one descends directly while the other is caught for many bounces among the pegs.

vector, z_t . To reason, the model takes its current state z_t , combines it with the input problem x , and applies a learned transformation Φ :

$$z_{t+1} = \Phi(z_t, x)$$

The machine repeats this loop— $z_0 \rightarrow z_1 \rightarrow z_2 \rightarrow \dots$ —updating its internal representation with every iteration. It continues until the state stops changing (or in practice, until state updates fall below a convergence threshold or hit a step budget), approaching an effective fixed point z^* where:

$$\Phi(z^*, x) = z^*$$

Once z^* is reached, further computation produces no new information. The machine

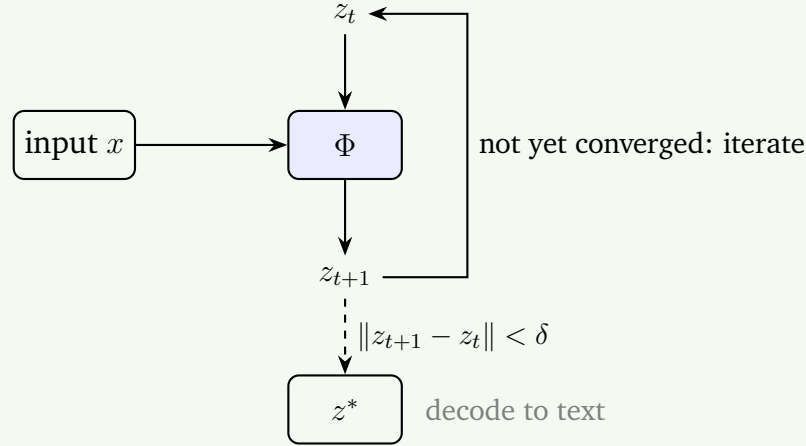


Figure 13.2: The recurrent-depth loop. The same map Φ is applied repeatedly to the latent state, with the problem x held fixed, until the update falls below a threshold δ (or a step budget is reached).

stops, translates z^* into readable text, and presents the solution. In this recurrent setup, “thinking time” is no longer a metaphor. It is the total number of iterations required for the dynamical system to settle into its fixed point. This raises a question: Why does a hard puzzle force this internal dynamical system to take so much longer to settle? A study by Lai et al. [1] mapped this invisible process. To see how the landscape changes with problem difficulty, the researchers kept the problem x and the network Φ completely fixed, varying only the model’s starting point in latent space, z_0 . Because the latent space contains hundreds or thousands of dimensions, they analyzed a two-dimensional slice through it, defined by a base point z_{base} and two orthogonal direction vectors u and v :

$$z_0(a, b) = z_{\text{base}} + a u + b v$$

By systematically sampling thousands of coordinate pairs (a, b) , the researchers dropped “chips” from thousands of slightly different initial conditions and counted how many iterations each point required to reach the fixed point z^* . They then color-coded each coordinate based on its convergence time. The resulting visualization revealed a striking structural shift. For simple problems, the map was smooth and predictable. Large, uniform regions of starting points all settled in roughly the same number of steps. But as problem difficulty increased, the smooth terrain fractured. The boundaries between fast-converging and slow-converging regions shattered into intricate, self-similar fractal patterns. Zooming in on a boundary revealed finer tendrils; zooming in further revealed even more complexity. Two initial states separated by a microscopic distance could yield drastically different trajectories: one might plunge straight to the answer in twenty steps, while its neighbor wandered through hundreds of iterations before reaching the exact same final destination. This fractal structure carries a mathematical signature. Consider

two initial states separated by a tiny perturbation ε . Let $f(\varepsilon)$ represent the fraction of such neighboring pairs that yield fundamentally different convergence times. In a smooth, well-behaved system, the boundary is a clean line, and uncertainty scales linearly with distance: $f(\varepsilon) \sim \varepsilon$. Measuring your starting position ten times more precisely reduces your uncertainty by a factor of ten. But in the presence of fractal basin boundaries, uncertainty obeys a power law governed by an uncertainty exponent, α :

$$f(\varepsilon) \sim \varepsilon^\alpha \quad \text{where } 0 < \alpha < 1$$

When α is small, uncertainty vanishes with agonizing slowness. For instance, if a hard reasoning task creates a landscape with $\alpha = 0.25$, refining your knowledge of the initial hidden state by a factor of ten reduces your uncertainty by only $10^{0.25} \approx 1.78$. No

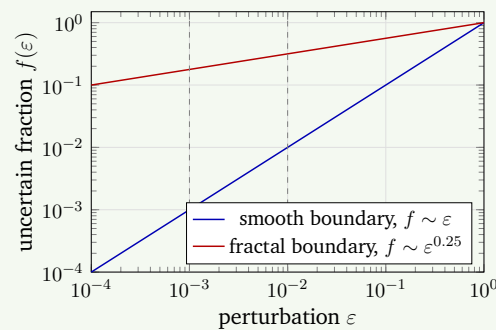


Figure 13.3: Uncertainty versus measurement precision on log-log axes. Refining ε by a factor of ten cuts f by ten on a smooth boundary but by only $10^{0.25} \approx 1.78$ when $\alpha = 0.25$.

matter how fine your resolution, the boundary remains stubbornly complex. To quantify how rapidly nearby trajectories diverge during these extended journeys, the researchers calculated the Fast Lyapunov Indicator (FLI), which measures the maximum rate of separation between neighboring trajectories over finite run times rather than infinite asymptotic limits. High FLI values formed bright, glowing networks along the fractal edges, providing direct evidence that the regions where the model is most sensitive to its initial state z_0 are precisely the areas where internal trajectories experience dynamic instability. What actually acts as the “pegs” inside the machine’s latent space? The authors discovered that long, chaotic trajectories occur because the system becomes temporarily trapped near saddle points. We know that a saddle point resembles a mountain pass. If you approach the pass from the valley floor, the surrounding terrain pulls you toward the center line. However, the pass is unstable in the cross direction: stray even slightly off-center, and you accelerate down the steep drop on either side. In the latent space of a reasoning model, these saddle points have a clear interpretation: they are almost-correct answers. When tasked with a complex problem, the model’s trajectory often drifts into a region representing a plausible half-truth: a Sudoku board with a single conflicting number, or a logical proof with a subtle flaw in step three. The saddle point acts as a temporary attractor, drawing the trajectory in. For dozens or hundreds of iterations, the state hovers near this near-miss. But because a saddle point is inherently unstable, the trajectory eventually finds the unstable direction, breaks free, and resumes its search for the true fixed point z^* . This phenomenon is known as transient chaos. The model is not permanently stuck in an infinite loop; it is temporarily lost in the vicinity of a compelling mistake. The study’s most revealing experiment tracked how this geometry emerges

during training. At initialization, the model’s workspace contains dozens of sinkholes that trap reasoning trajectories before they can ever find the right path. Many of these sinks represent wrong answers. Early in training, if a model encounters a hard problem, it simply falls into one of these wrong sinks and stops, confidently outputting an incorrect answer because it reached a stable state where $\Phi(z, x) = z$. As training progresses, the optimization algorithm destabilizes these incorrect fixed points, transforming them from sinks into saddles. This structural shift alters how the model processes errors. The

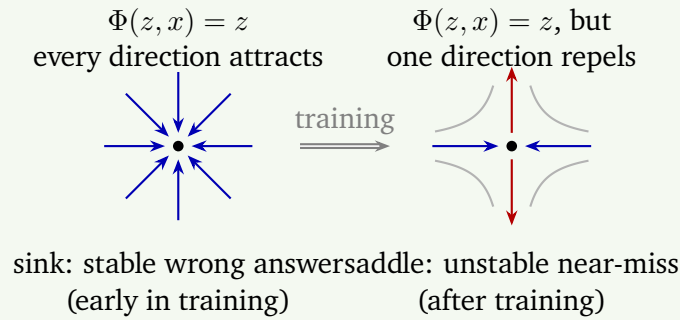


Figure 13.4: Training destabilizes incorrect fixed points. A sink traps every trajectory permanently; a saddle draws trajectories in along its stable direction, holds them for a while, then expels them along its unstable direction. That escape is what produces transient chaos.

machine learns that a flawed answer must not be a comfortable resting place. It must approach an error, recognize through its internal dynamics that the state is invalid, and push itself away. However, this capability comes at a computational cost. By destroying wrong stable answers and turning them into unstable saddles, the training process generates transient chaos. The mechanism that enables the model to escape wrong answers is the same mechanism that extends the time required to solve hard problems. The structural complexity of the landscape is the trade-off required to avoid persistent errors. Ultimately, this research demonstrates that computational difficulty possesses a physical signature inside learned dynamical systems. “Thinking harder” is not merely performing additional arithmetic; it is navigating a geometric space containing unstable traps, plausible wrong turns, and fractal delays. While these findings apply specifically to recurrent-depth models rather than standard feedforward architectures, they provide a precise framework for understanding automated reasoning. Hard problems require a landscape where wrong answers are easy to approach, but unstable enough to force eventual escape. Every initial state z_0 ultimately falls into the single slot z^* ; what varies is how long the trajectory spends caught among the pegs of the model’s own near-misses.

References

- [1] Lai et al., “Transient Chaos and Fractal Boundaries in Recurrent Reasoning Models,” *arXiv preprint arXiv:2609.04963*, 2026.

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And to you, the reader: thank you for picking this up. We hope you enjoy it as much as we enjoyed making it.

— *The Editors, Elements*

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